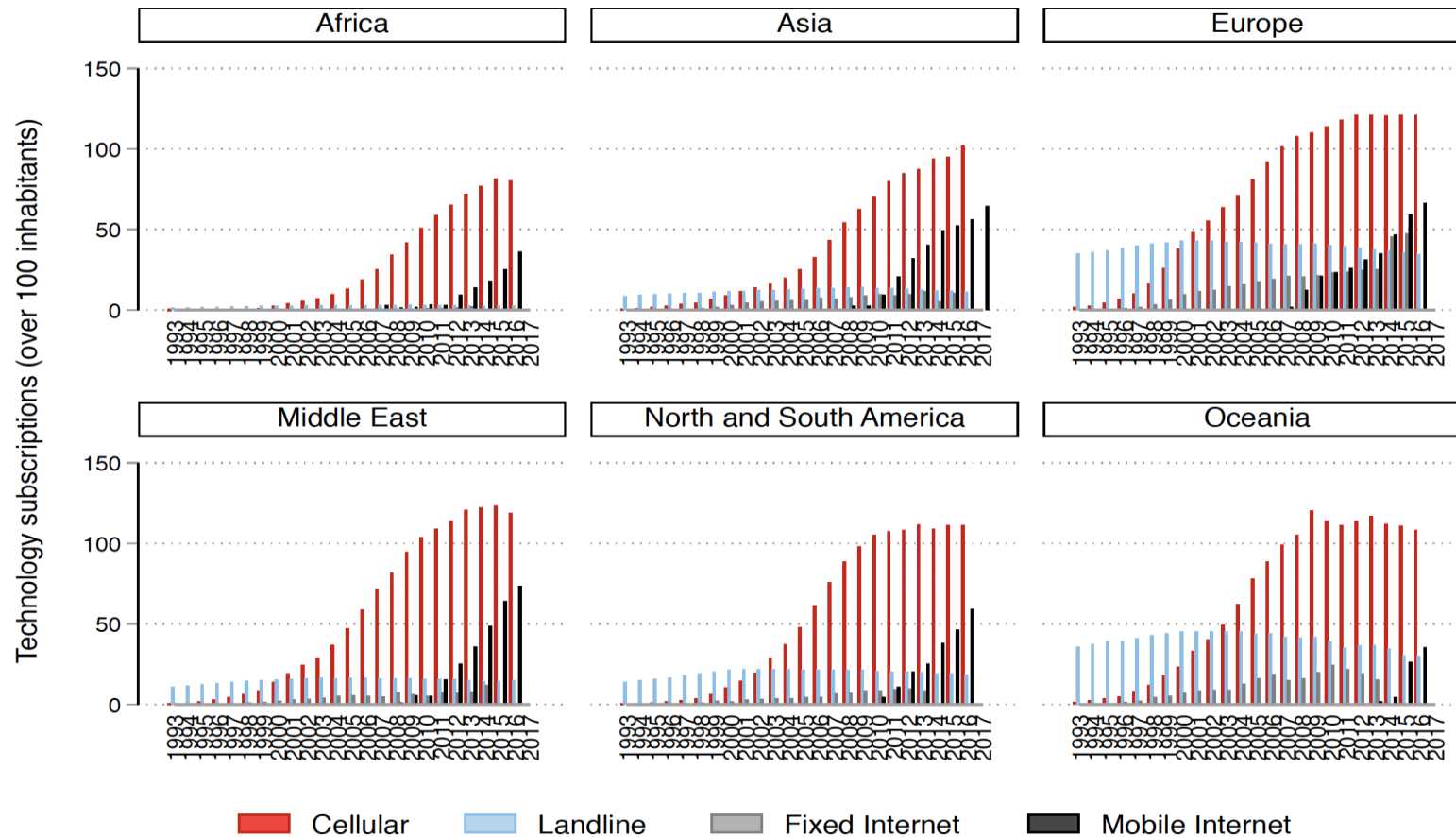


Gender Inequalities in a Digital World: Insights from the Digital Gender Gaps Project

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Nuffield College
Leverhulme Centre for Demographic Science
University of Oxford

September 15, 2025
SICSS, Saarland Uni, Saarbrücken

The Digital Revolution



Technology subscriptions per capita, by region, 1993-2017.

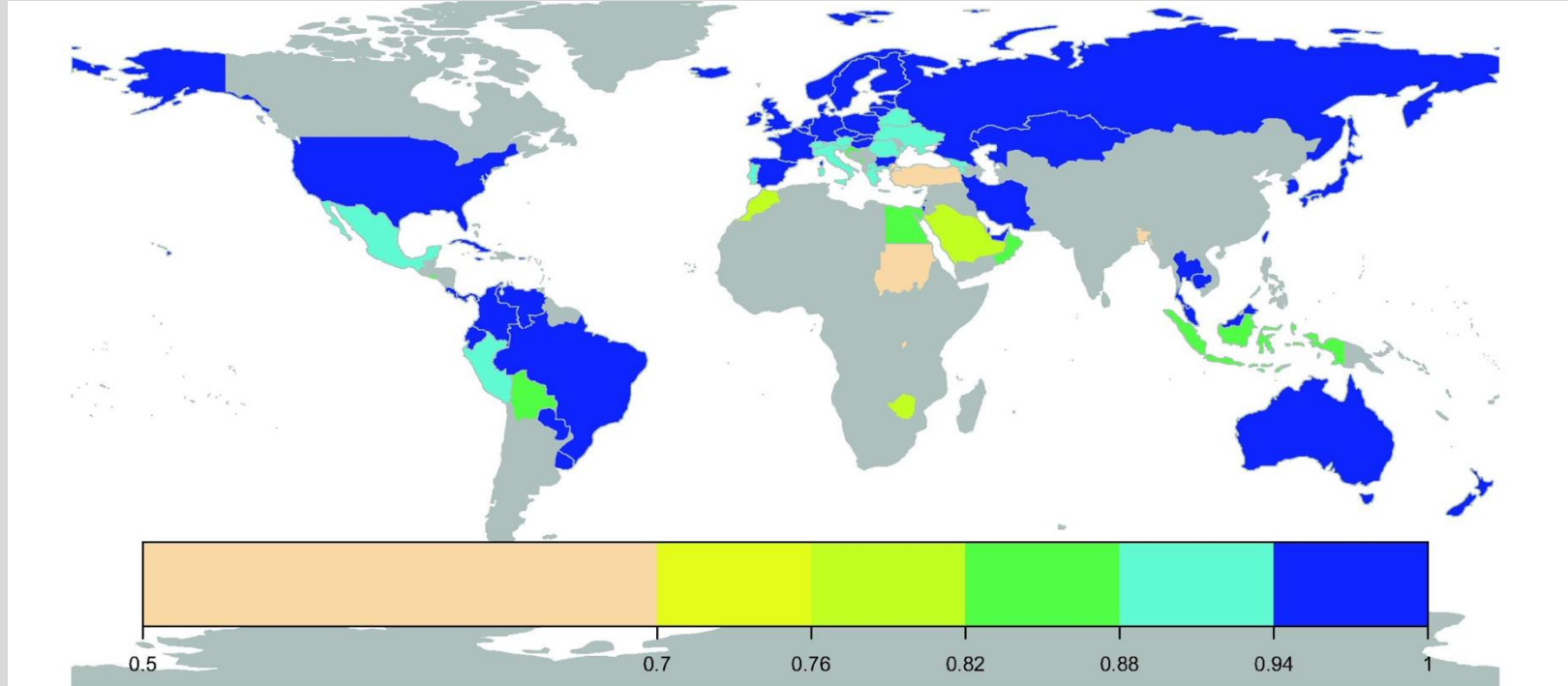
Data from International Telecommunications Union (ITU)

The Digital Revolution

- The digital revolution is:
 - **Social revolution:**
 - SDG 5 on gender equality pledges to “enhance the use of...information and communication technologies to promote the empowerment of women.”
 - What is the empirical evidence? How, for whom, under what conditions?
 - Theoretically ambiguous effects – can empower the marginalised but also reinforce or exacerbate inequalities
 - **Data revolution:**

Our use of new technologies such as mobile phones, web and social media generates new data streams that have the potential to provide real-time information on SDGs

The Data Gap

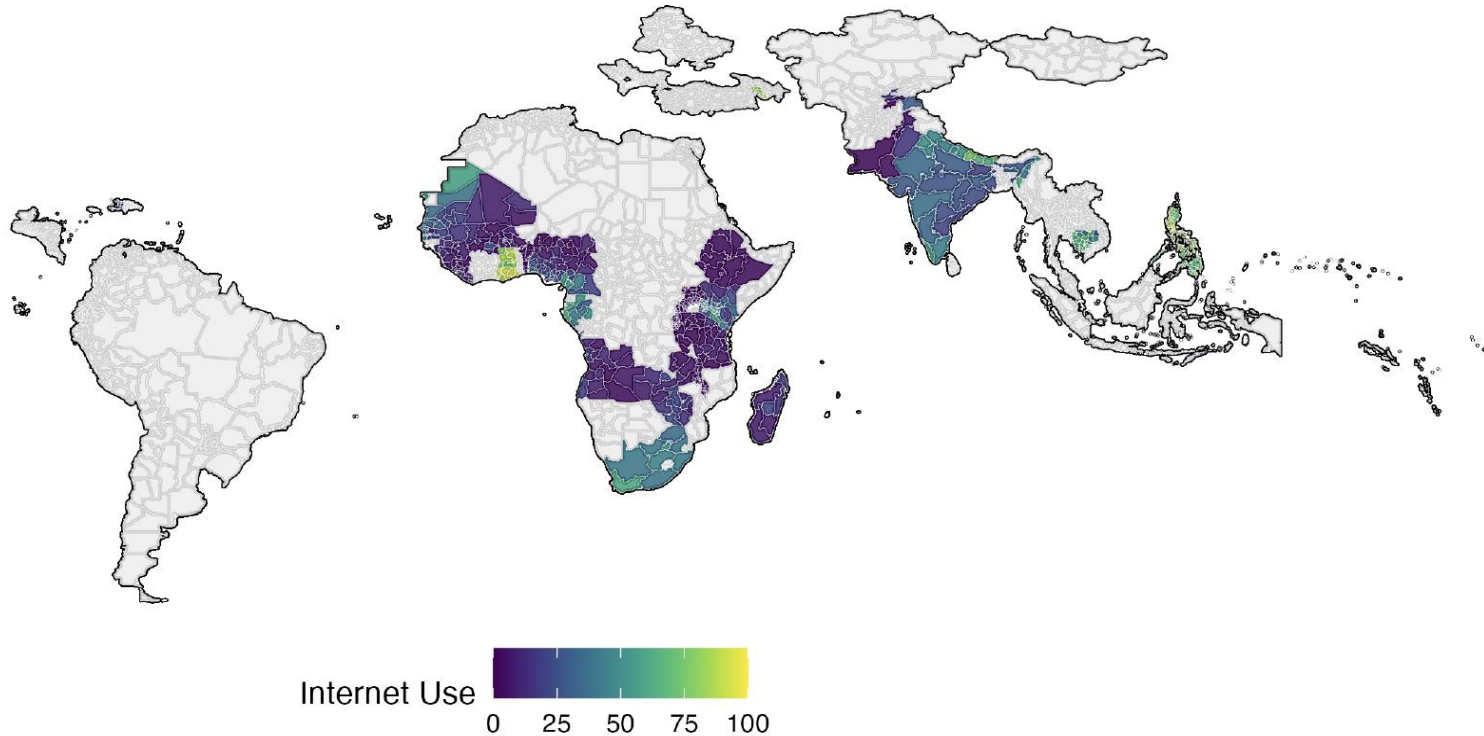


(a) Internet Gender Gap Index computed using ITU ground truth data

Fatehkia, Masoomali, Ridhi Kashyap, and Ingmar Weber. "Using Facebook ad data to track the global digital gender gap." *World Development* 107 (2018): 189-209.

The Data Gap

Women, observed



Women who have used the internet at subnational resolution (admin-1) computed using Demographic and Health Surveys.

The Digital Revolution



- The digital revolution is:
 - **Social revolution:**
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 - What is the empirical evidence? How, for whom, under what conditions?
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Our use of the web, social media and mobile phones generates new data streams that have the potential to provide real-time information on digital connectivity and behaviours

Digital Gender Gaps Project



- The Digital Gender Gaps (DGG) project seeks to **deepen data and evidence** on:
 - (1) **Gender gaps in digital connectivity and behaviours** at national- and subnational-levels, by integrating social media, geospatial data and population survey and census datasets;

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Digital Gender Gaps Project

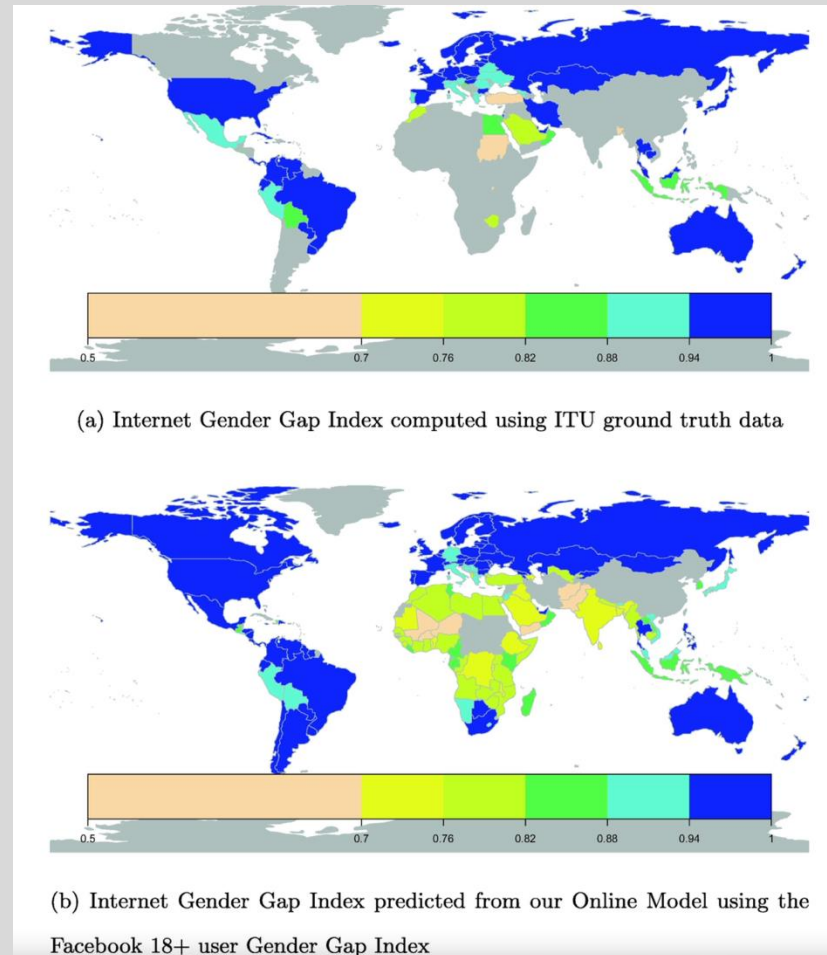


(1) Gender gaps in digital connectivity and behaviours at national- and subnational-levels

Applications:

- Predicting gender gap indicators computed using social media advertising data
 - Internet and mobile access gender gaps (www.digitalgendergaps.org, using Facebook)
 - Professional gender gaps (using LinkedIn)
- Expanding geographical coverage, more timely data on social indicators and for monitoring rapidly evolving situations

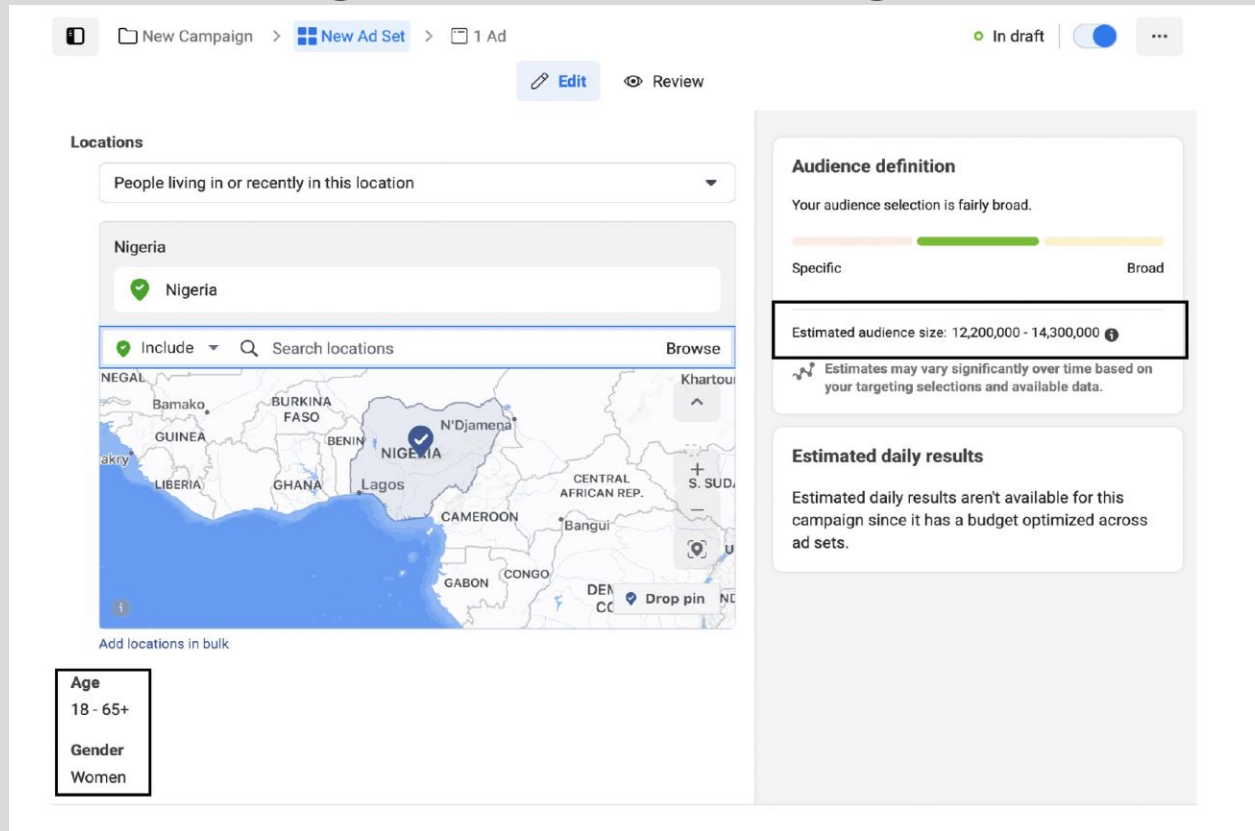
Internet GGI predicted using FB GGI



Fatehkia, Masoomali, Ridhi Kashyap, and Ingmar Weber. "Using Facebook ad data to track the global digital gender gap." *World Development* 107 (2018): 189-209.

Facebook Marketing API

- Analysing global gender gaps in digital connectivity using **social media data advertising data validated against population surveys**

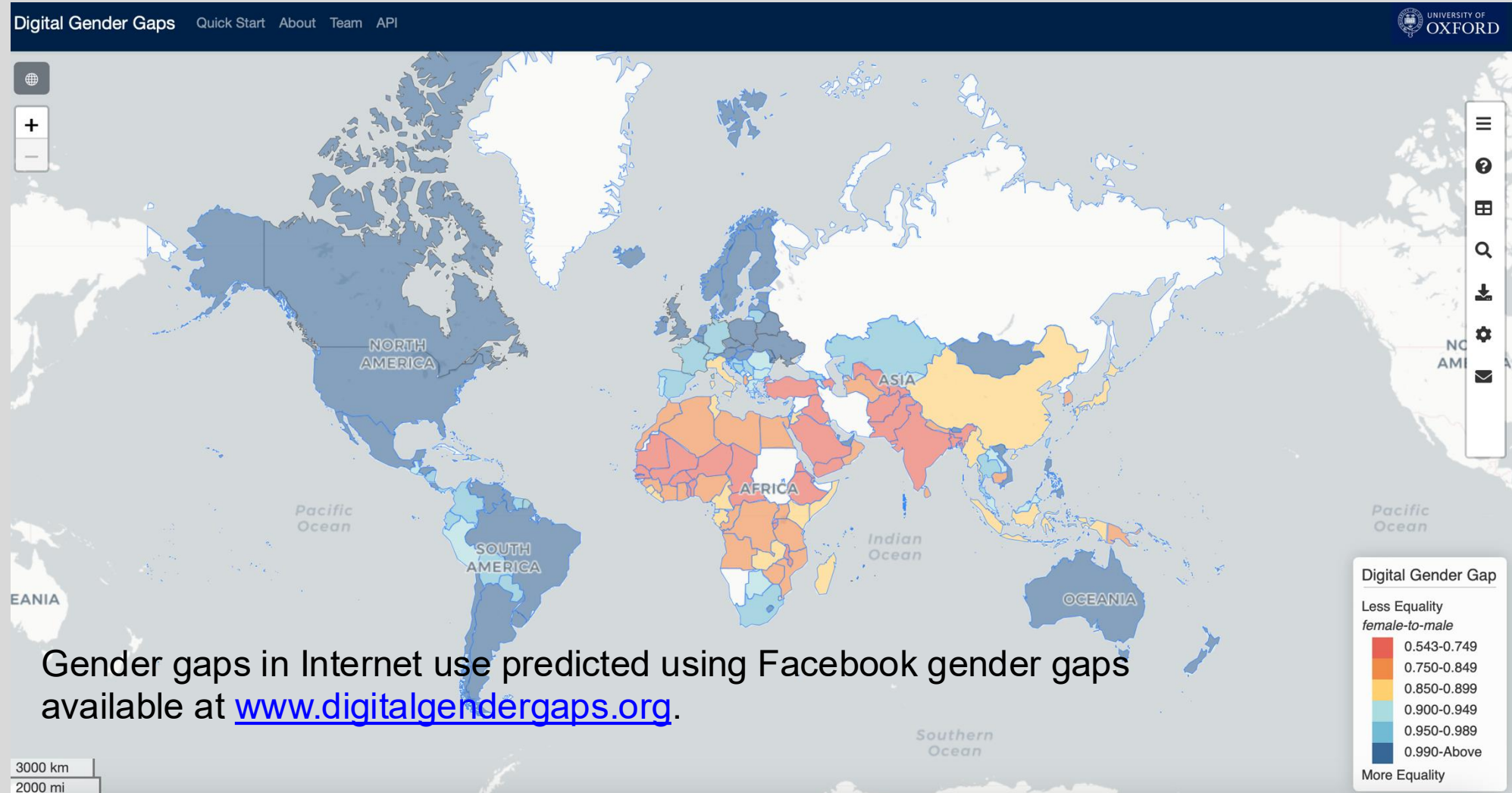


- How many users of 'x' characteristics (gender, location, age, device type, etc) are on a given platform?

Facebook Ads Manager:

<https://www.facebook.com/adsmanager>

Internet GGI predicted using FB GGI



Facebook Gender Gap Index

$$FB\ GGI = \frac{\textit{Female to male ratio of Facebook users}}{\textit{Female to male ratio of the population}}$$

- Using ITU data, we compute a ground truth internet gender gap indicator derived from survey data:

- $$\textit{Internet GGI} = \frac{\% \textit{ of female population using the Internet}}{\% \textit{ of male population using the Internet}}$$

- Similarly, we compute Mobile GGI

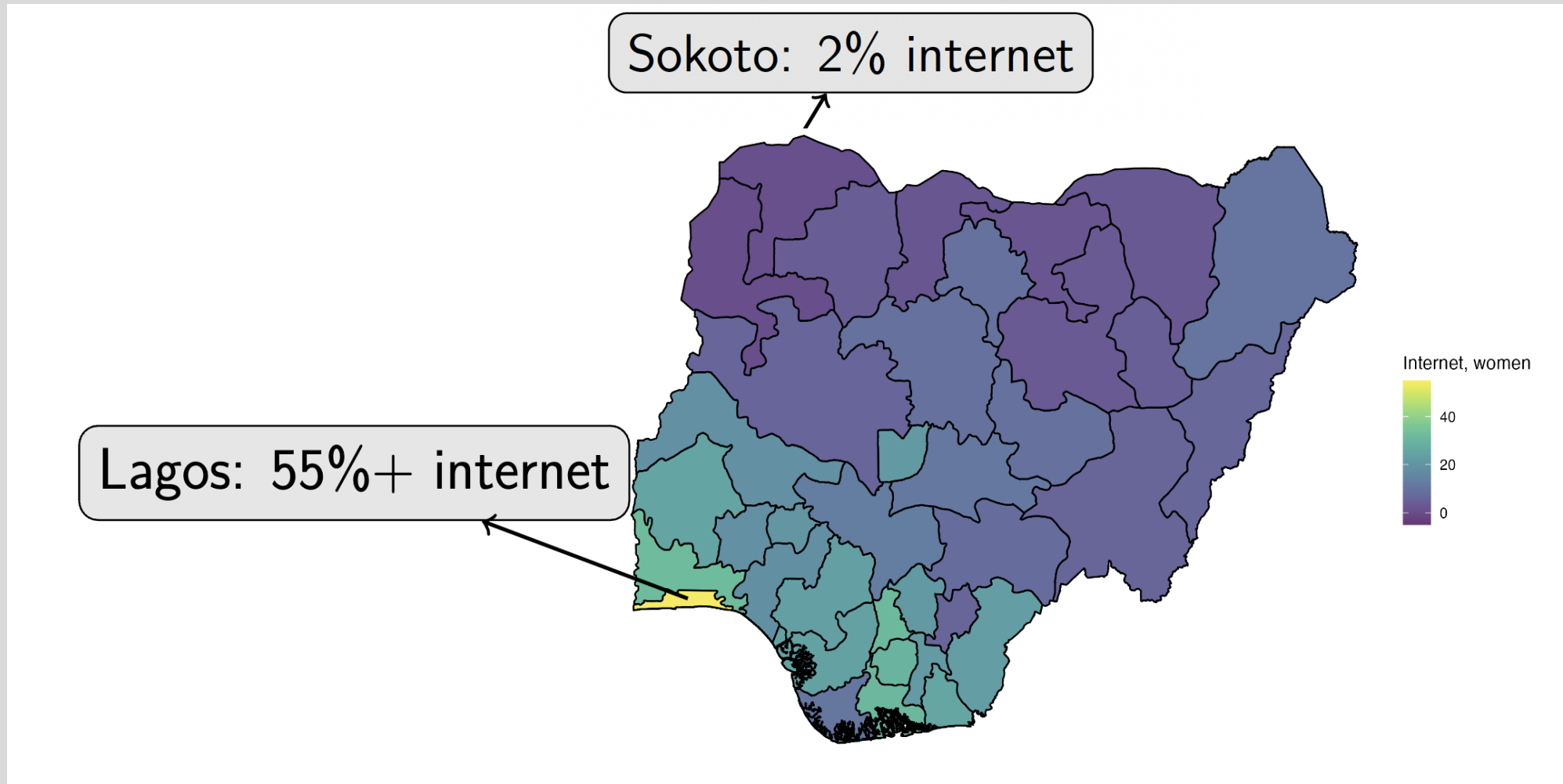
Facebook Gender Gap Index



FB GGI $\approx$ Internet GGI

- Correlation of FB GGI 18+ with Internet GGI is ~ 0.8 , mobile GGI ~ 0.7
- Stronger than any other development indicator (including internet penetration)
- We fit three types of predictive models:
 - online model
 - online-offline model
 - offline model

Subnational Variation

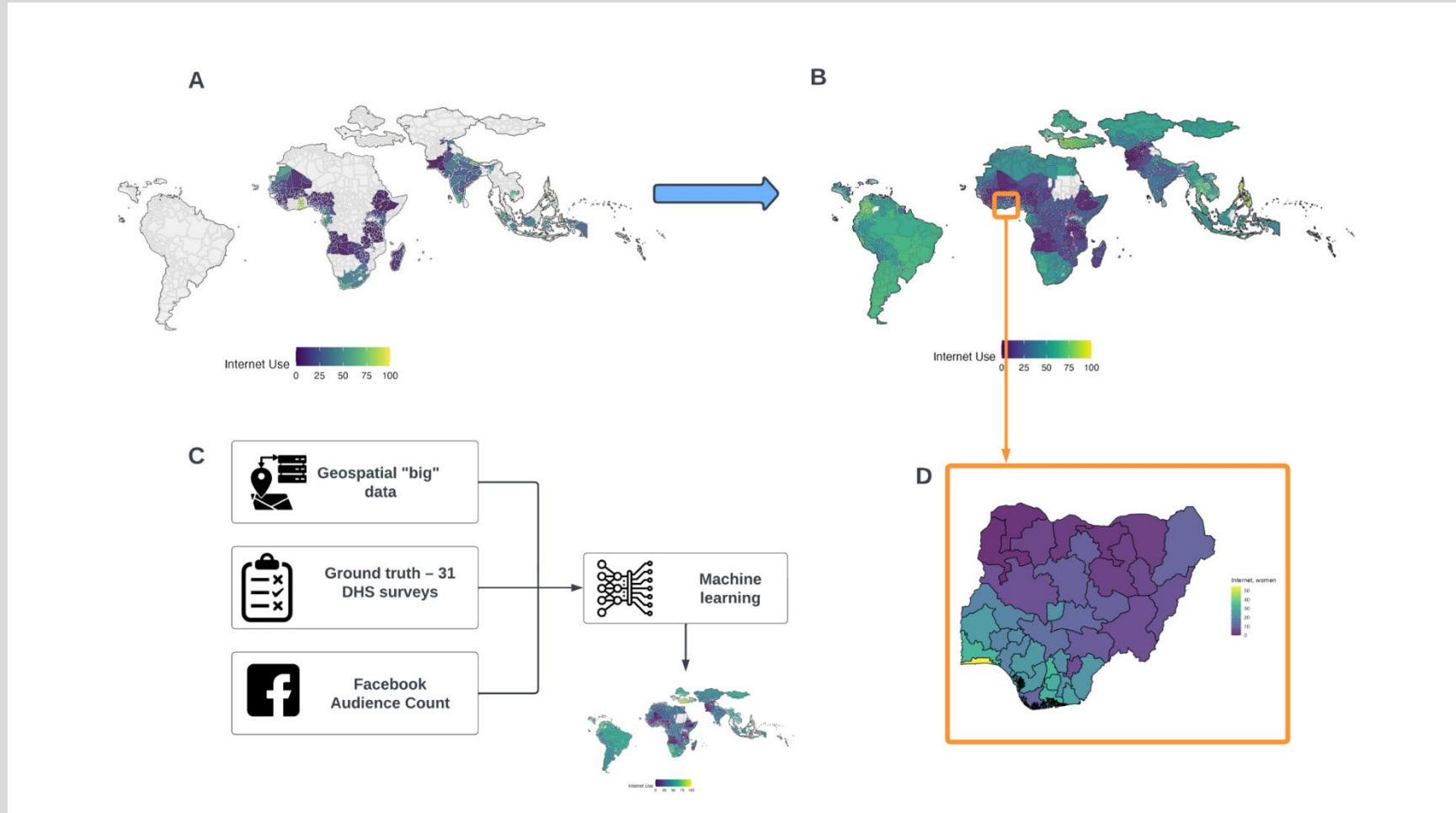


Internet use (last twelve months), women, Nigeria. From Demographic and Health Surveys (DHS).

Subnational Estimates of Digital Gender Gaps

- **Goal:** develop subnational estimates of digital adoption and gender gaps for LMICs
- GADM-1 resolution, 117 countries, 2,075 regions
- **Indicators:** gender-disaggregated internet use and mobile ownership (% internet user, women/men, %mobile owner, women/men) and gender gaps (female-to-male ratio)

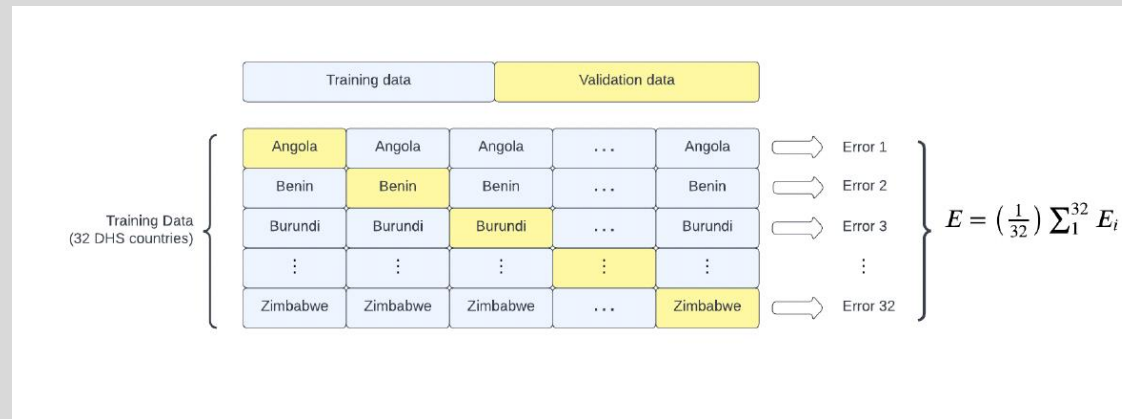
Our approach



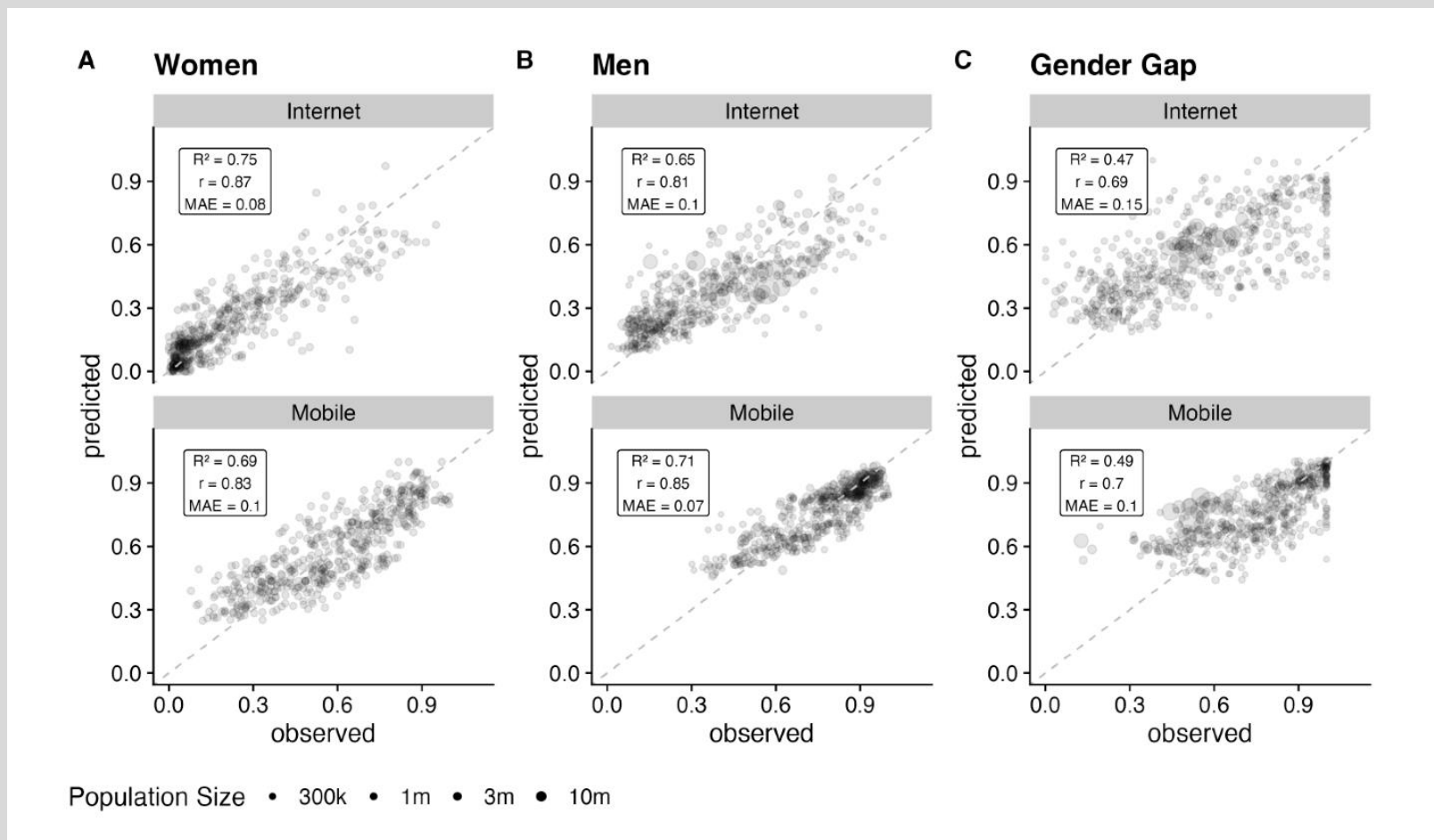
Panel A shows the 33 countries for which ground truth data are available at subnational resolution. **Panel C** shows inputs to the ML model used to predict digital indicators for 117 LMICs. **Panel B** shows model-based estimates of internet adoption by women. **Panel D** shows enlarged estimates for Nigeria.

- **Ground truth:** Demographic and Health Surveys, representative at GADM-1 level, 2016-22 covering 33 countries
- **‘Online’ predictors:** Facebook penetration (proportion Facebook users in subnational area by gender, age, device type), or gender gaps, using Facebook marketing API
- **‘Offline’ predictors:**
 - Satellite-derived nightlight data
 - Population density (WorldPop)
 - Subnational education index, income index, human development index (HDI), gender development index (GDI)

- Ensemble Superlearner: best weighted combination of ML algorithms
- To assess predictive performance, three validation strategies:
 - Conventional 10-fold cross-validation
 - Leave-one-country-out cross-validation (LOCO-CV)
 - External benchmarking against Living Standards Measurement Studies (LSMS) and Multiple Indicator Cluster Surveys (MICS)

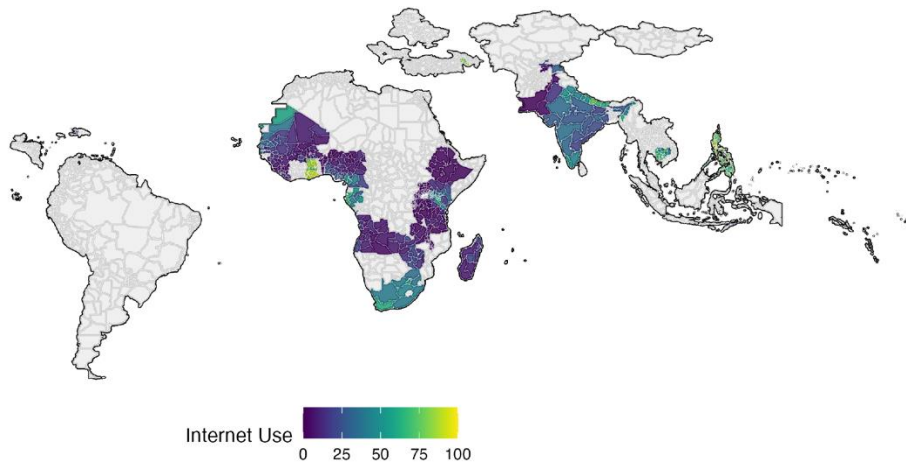


Overall predictive accuracy

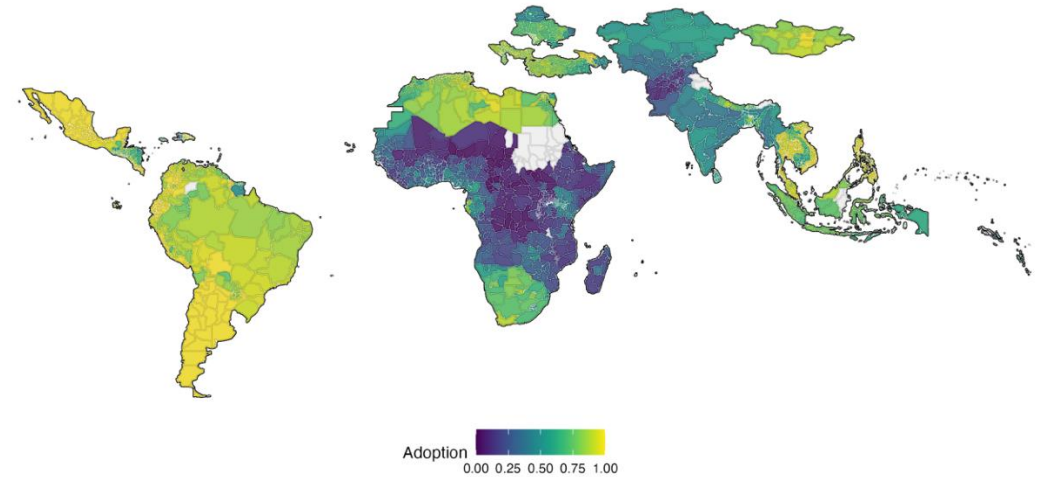


Expanded geographical coverage

Women, observed



Internet Use (Women, Predicted)



Improved temporal resolution

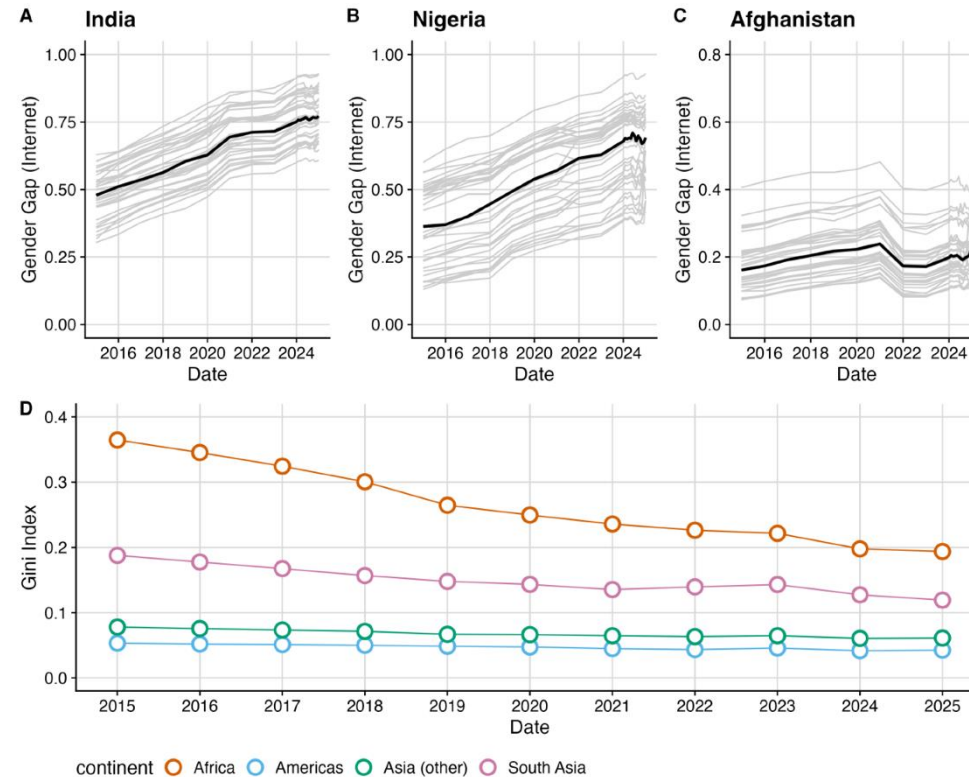
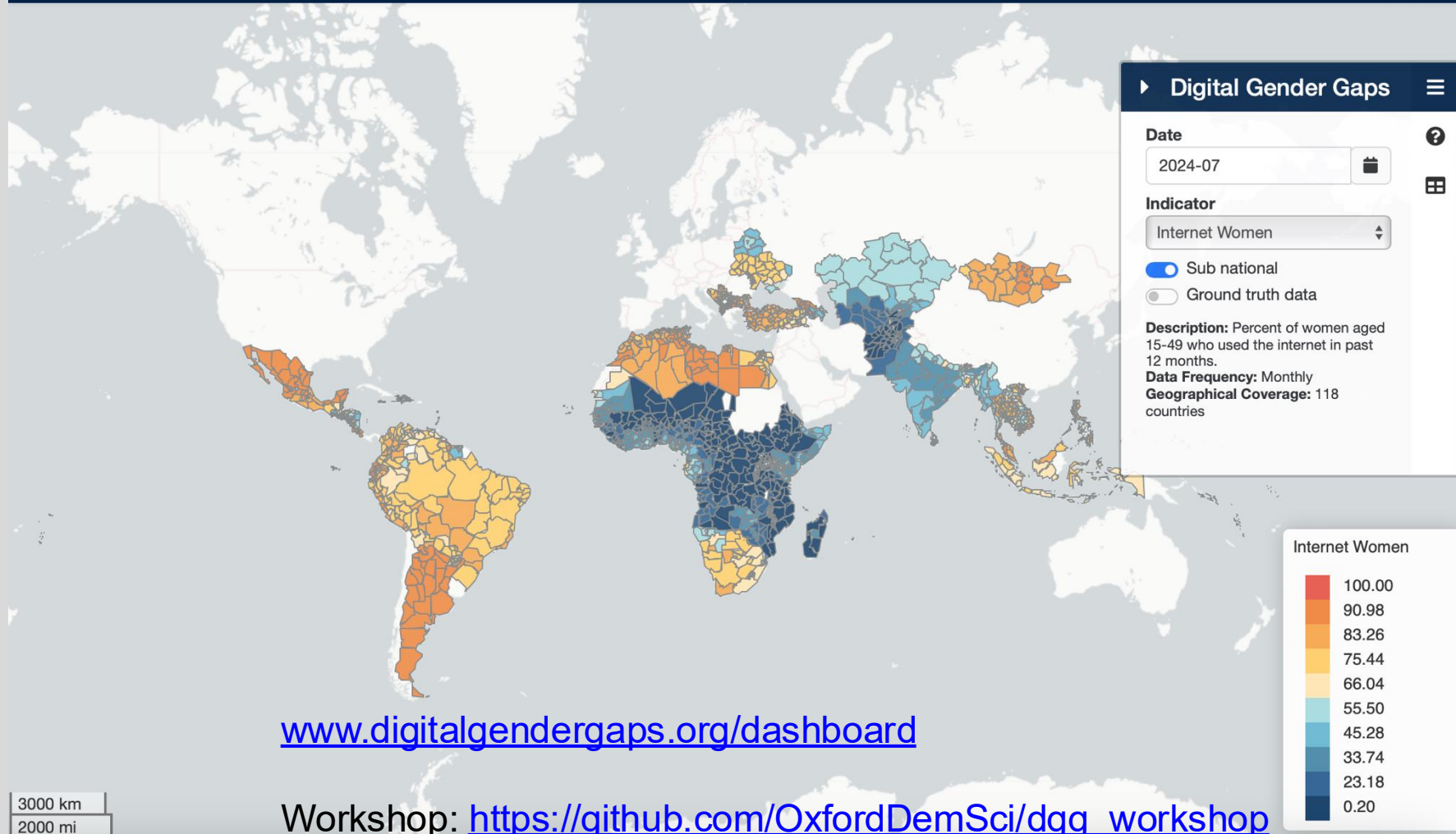


Figure 7: (A–C) Trends over time in India, Nigeria, and Afghanistan. The light grey lines show subnational change over time, and the dark shaded line shows the median trend for each country. (D) Change in within-country inequality for the internet gender gap, as measured by the subnational GINI index.

Interactive Dashboard

Digital Gender Gaps



LinkedIn Ad Audience Estimates

**Use a matched audience** (optional)
Custom targeting options to reach your website visitors, contacts, and target accounts. [Select](#)

Target by

What location do you want to target? (required)

include

Start typing a country, state, city, or town...

See full list

include

Netherlands

☒ Target people who permanently live or work in the selected location(s).
Deliver ads to people who reside in the selected **location(s)** and are not recent visitors

What gender do you want to target?

☐ All

☐ Female

☒ Male

Select specific targeting criteria to zero in on your ideal audience:

Company name	Company industry	Company size	Job title	Job function
Job seniority	Member schools	Fields of study	Degrees	Member skills

Your estimated target audience
4,500,000+ LinkedIn members

[See more](#)

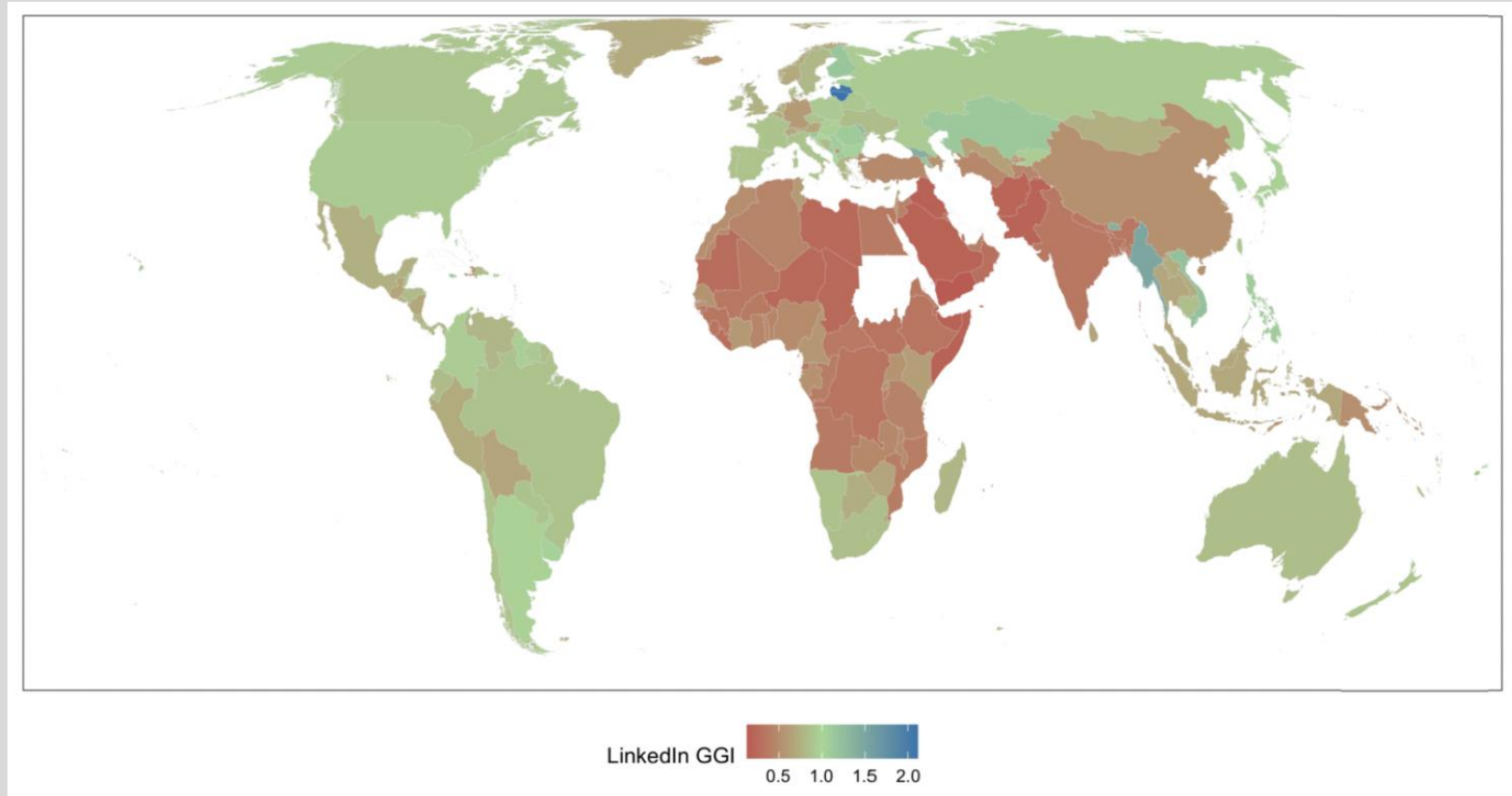
Netherlands

Male

Audience expansion: Enabled

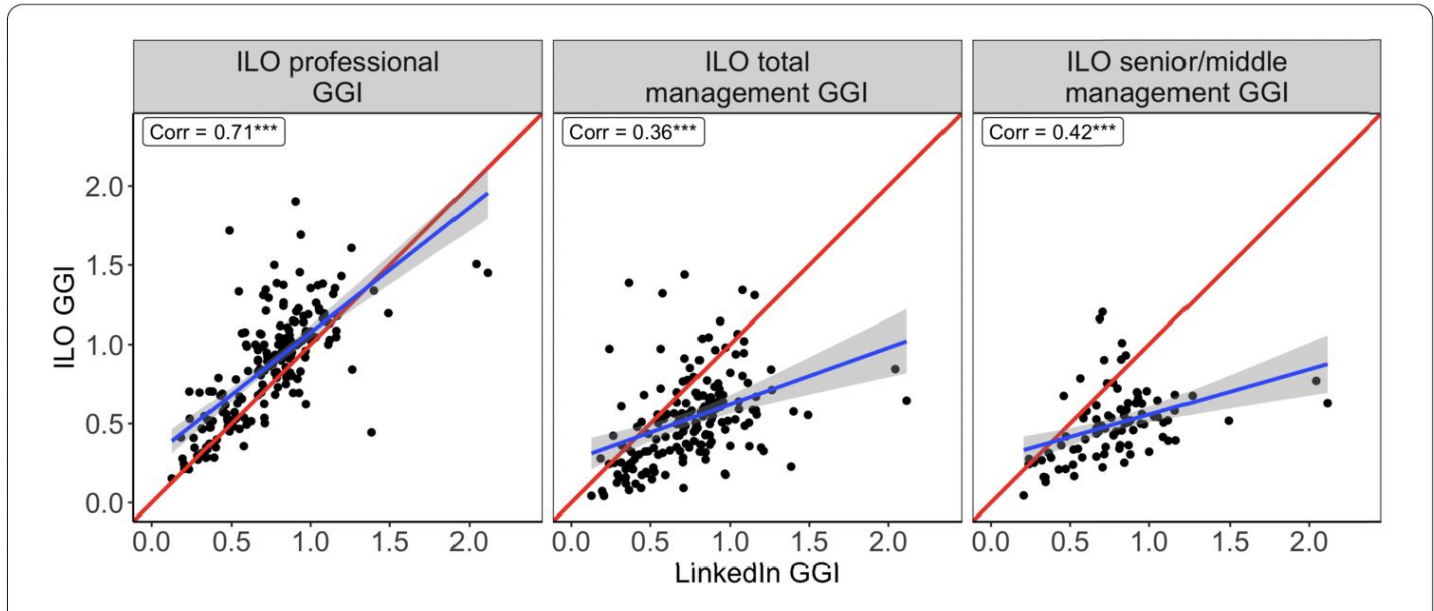
LinkedIn tools may not be used to discriminate based on personal characteristics like gender, age, or actual or perceived race/ethnicity. [Learn more](#)

LinkedIn Gender Gap Index



LinkedIn Gender Gap Index (female-to-male ratio of LinkedIn users by country) across the world, computed using LinkedIn advertising data

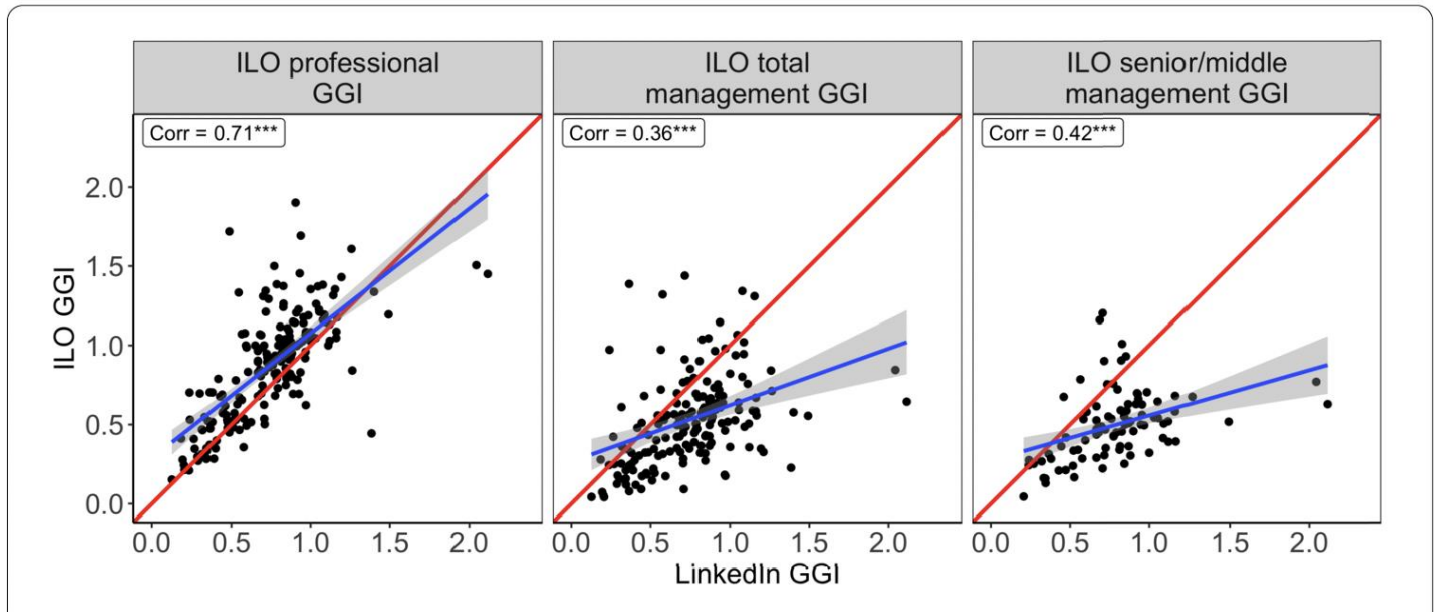
Professional Gender Gaps



Predicts ILO Professional Gender Gap Index well

LinkedIn Gender Gap Index (female-to-male ratio) and International Labour Organisation (ILO) Gender Gap Index across countries

Professional Gender Gaps



LinkedIn Gender Gap Index (female-to-male ratio) and International Labour Organisation (ILO) Gender Gap Index across countries

Management gender gaps less well → Women in management or senior/middle management roles are *overrepresented* on LinkedIn relative to labour force: online networking may provide greater benefits

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Two Examples

- 1) How does **online professional networking and outreach** via social media impact on women's professional outcomes? Can these tools mitigate gender inequalities in professional labour markets?
 - Information technology professionals and LinkedIn in the UK and US (with Ghazal Kalhor, Hannah Gardner, Ingmar Weber)

- 2) How does the **spread of mobile internet** via 3G networks affect women's status in the household and demographic behaviours?
 - 3G expansion, experiences of intimate partner violence and fertility behaviours in Nigeria (with Prashant Poddar, Casey Breen, Till Koebe, Valentina Rotondi)

Potential Impacts

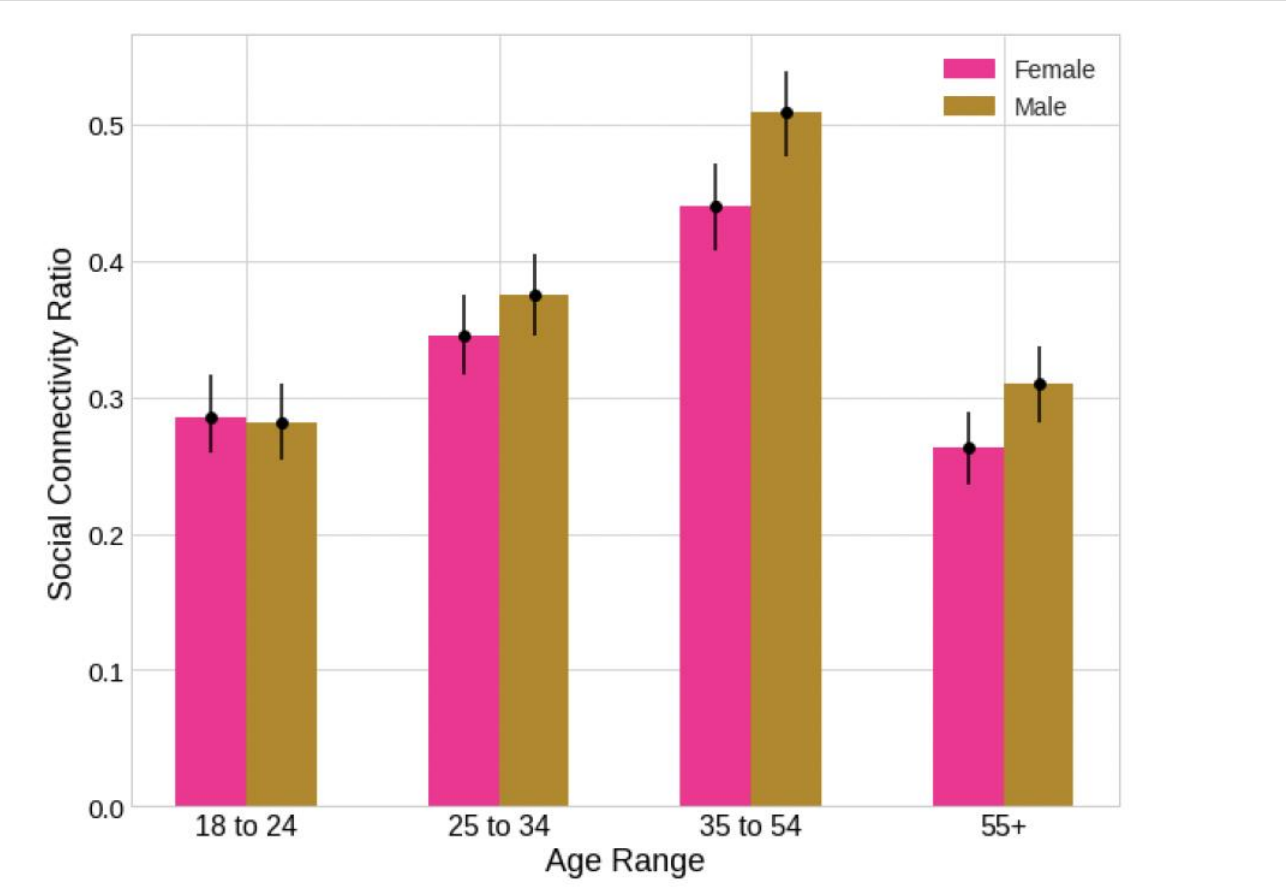
Social media can **increase visibility** and **provide opportunities** for more flexible forms of networking

- Greater accessibility for those with more constraints, e.g. care
- Lower cost and time-efficient
- Wider public reach

Social media can **reinforce** existing gender disparities

- Start from a “non-neutral” position
- Offline Networks reinforce online inequality
- Feminine modesty effect: self-promote less

Gender Gaps in Online Professional Networking



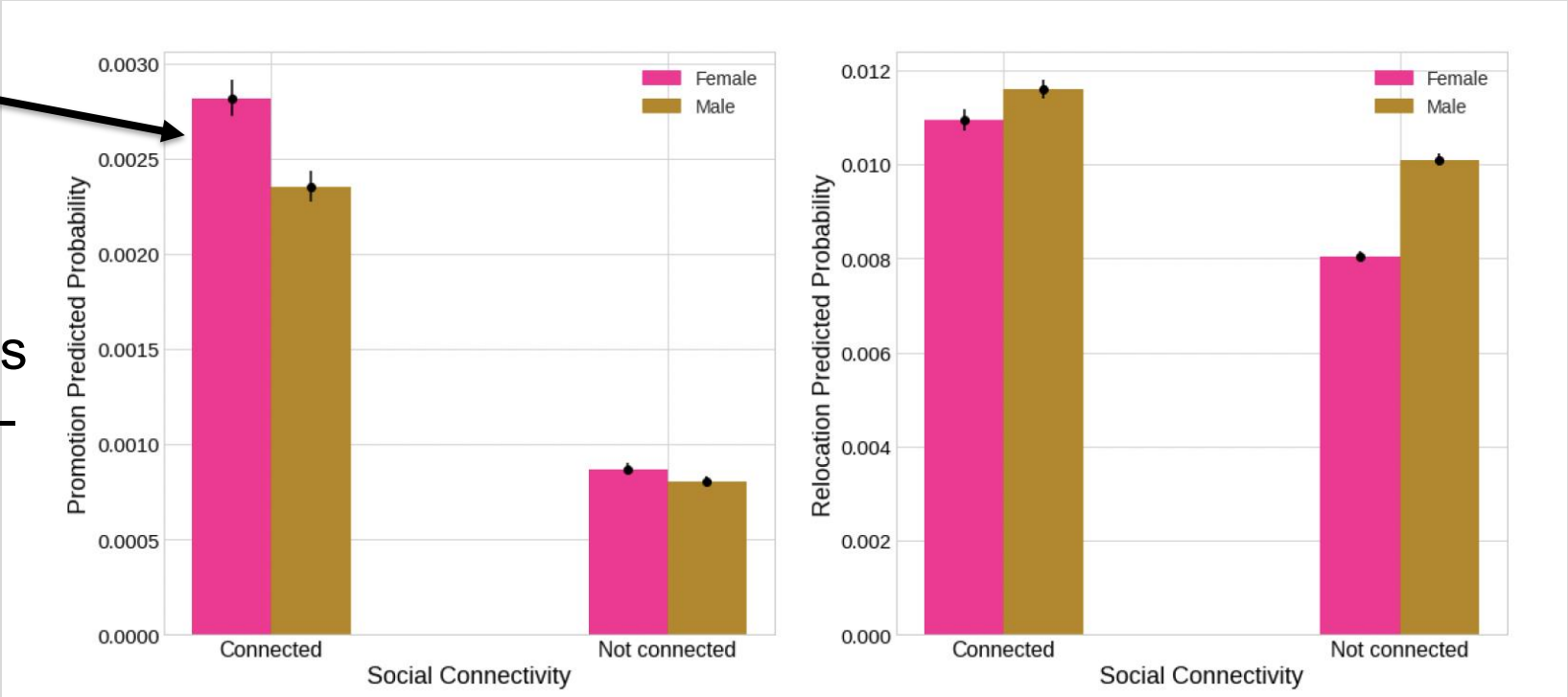
Women have less advantageous online professional networks on LinkedIn

Proportion of LinkedIn users working in information and technology industry, connected to those working in big-five tech firms, UK and US.

Gender Gaps in Online Professional Networking

How is online social connectivity associated with promotion and relocation outcomes?

Those
'connected' to
high status firms
more likely to
report promotions
and relocations –
with larger
payoffs for
women



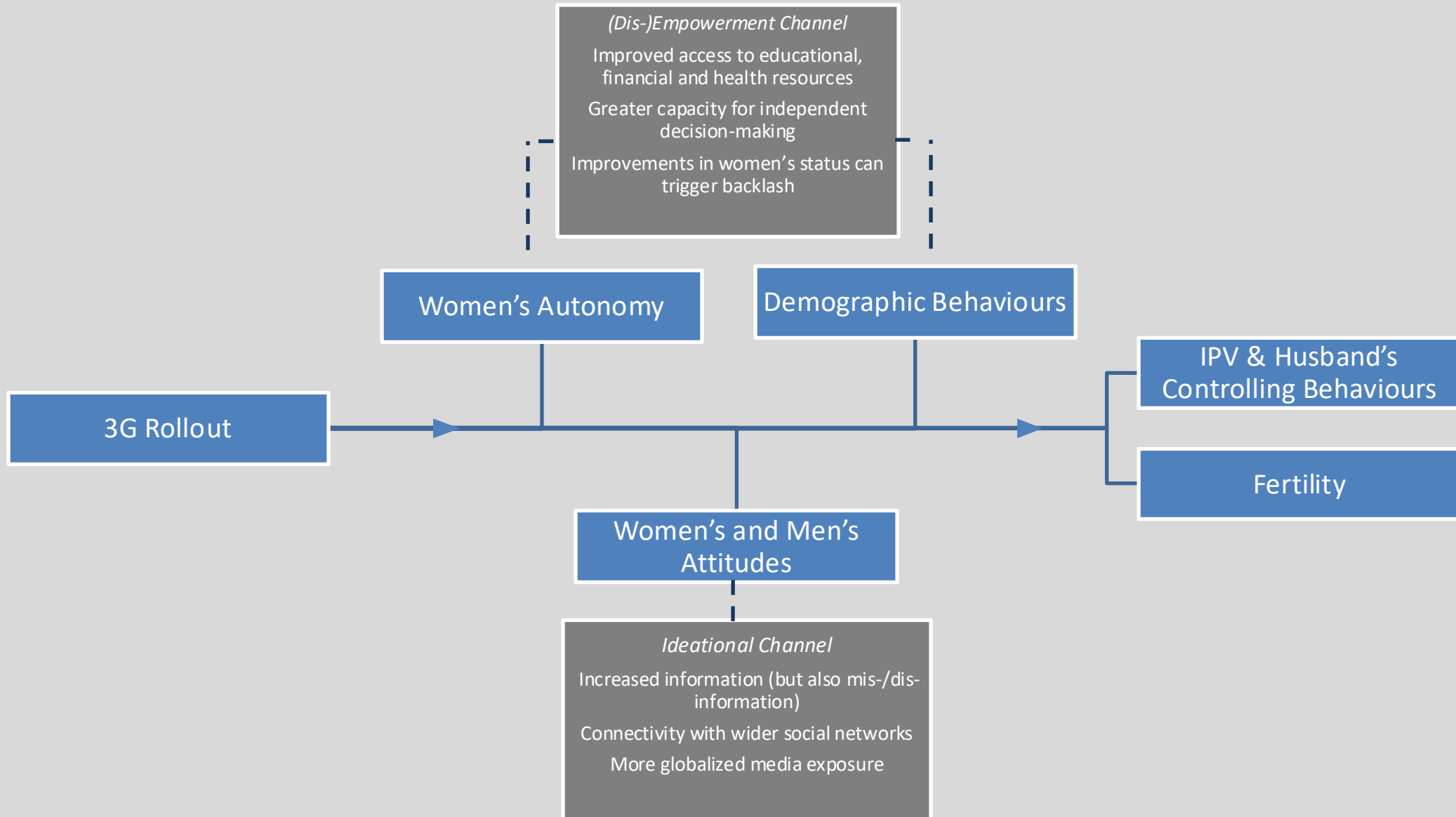
Research Question

- How does the spread of 3G+ mobile networks affect women's status in the household?
- Experience of intimate partner violence (IPV) and fertility: key markers of women's status
- Nigeria
 - Context of rapidly expanding 3G networks as enabler of internet access with limited fixed internet infrastructure
 - Patriarchal norms with high prevalence of IPV
 - Total fertility rate: 5.7 children per women as per 2018 DHS, from 6.1 in 2008, characterized by high and slow reductions in TFR
 - Access to high quality mobile coverage maps over time

Technology impacts: prior pathways

- Mass media technologies can spread of new ideas and norms (Barber and Axinn 2004)
- Diffusion theories of fertility decline: technology can alter paths for social learning and social interaction, which can shift gender roles and demographic behaviours (Montgomery and Casterline 1996)
- TV: exposure to new information – impact on fertility ideals, son preference and women's status in the household (e.g. Jensen and Oster 2009, La Ferrara et al 2012)

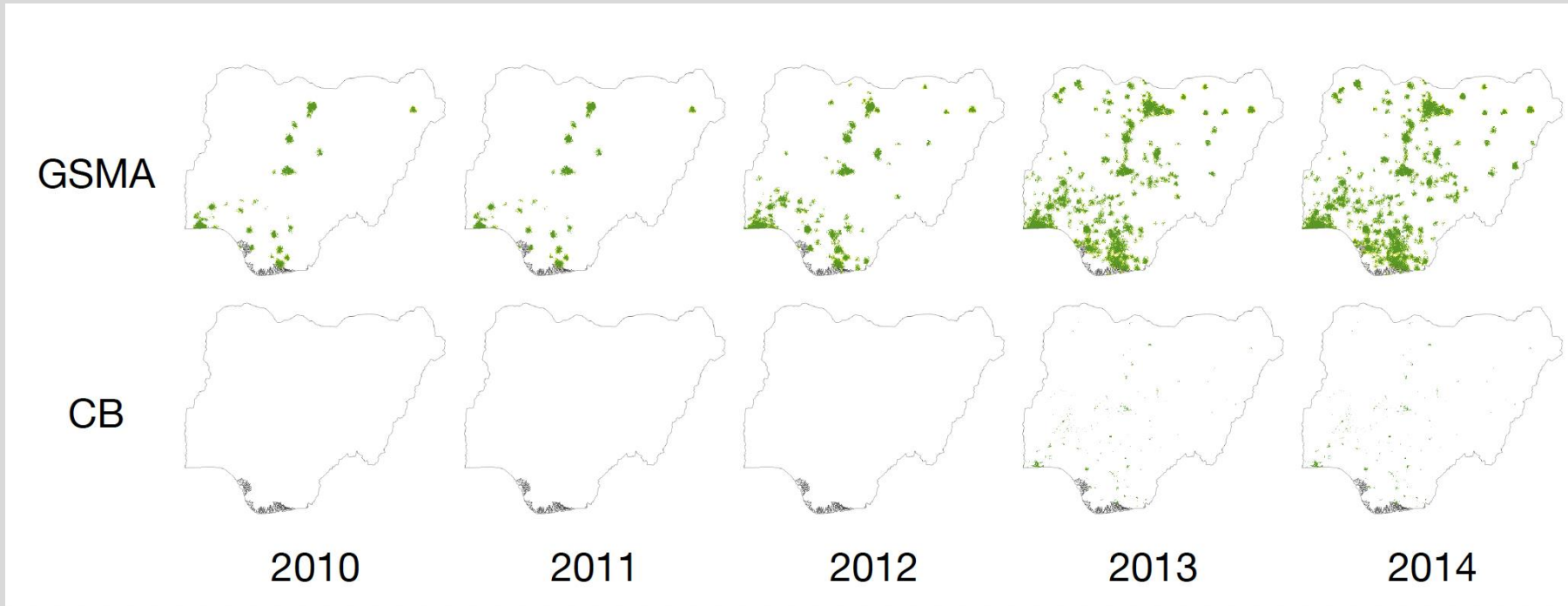
3G Internet: Channels



3G rollout

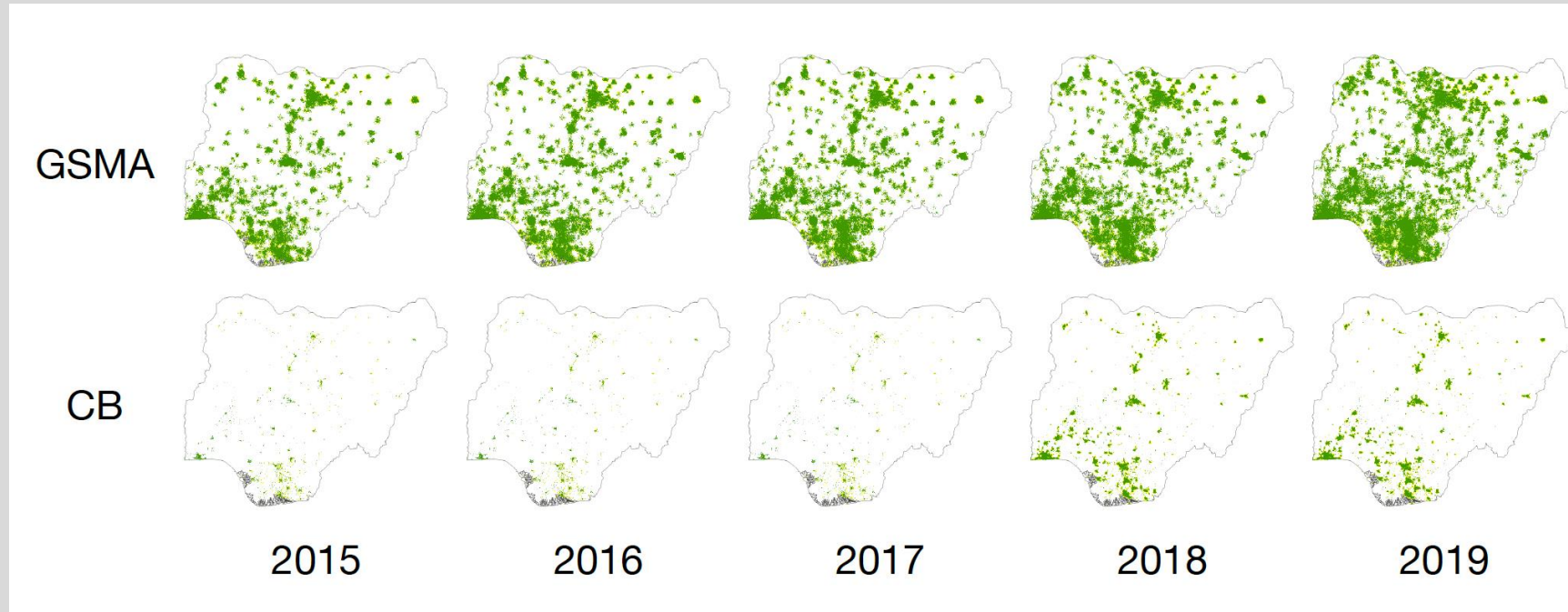
- We study the impacts of 3G rollout on women's experience of intimate partner violence (IPV) and fertility
- We link mobile coverage maps (from GSMA) to geo-referenced Demographic and Health Survey Data from Nigeria
 - Improvements in data quality of mobile coverages

Nigeria 3G Rollout



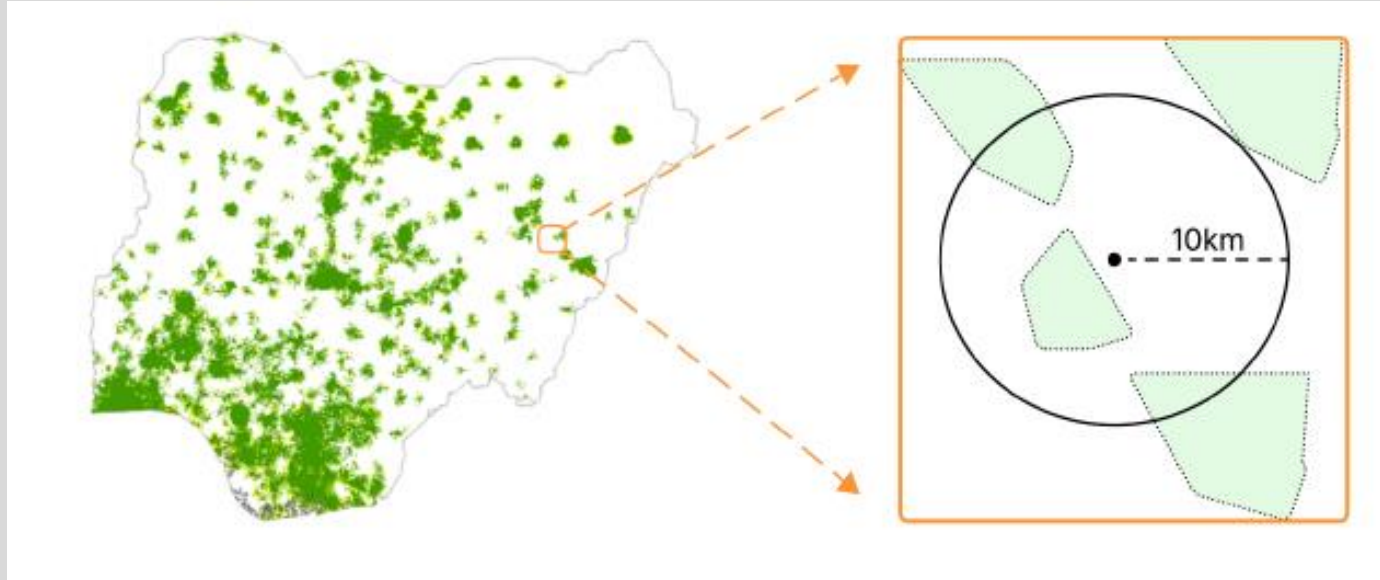
Mobile coverage rollout comparing GSMA and Collins Bartholomews (CB) coverage maps

Nigeria 3G Rollout



Mobile coverage rollout comparing GSMA and Collins Bartholomews (CB) coverage maps

Nigeria 3G Rollout



Computing population-weighted share of 3G coverage by creating a 10km (rural areas) or 2 km (urban areas) buffer around each DHS cluster

Identification Design

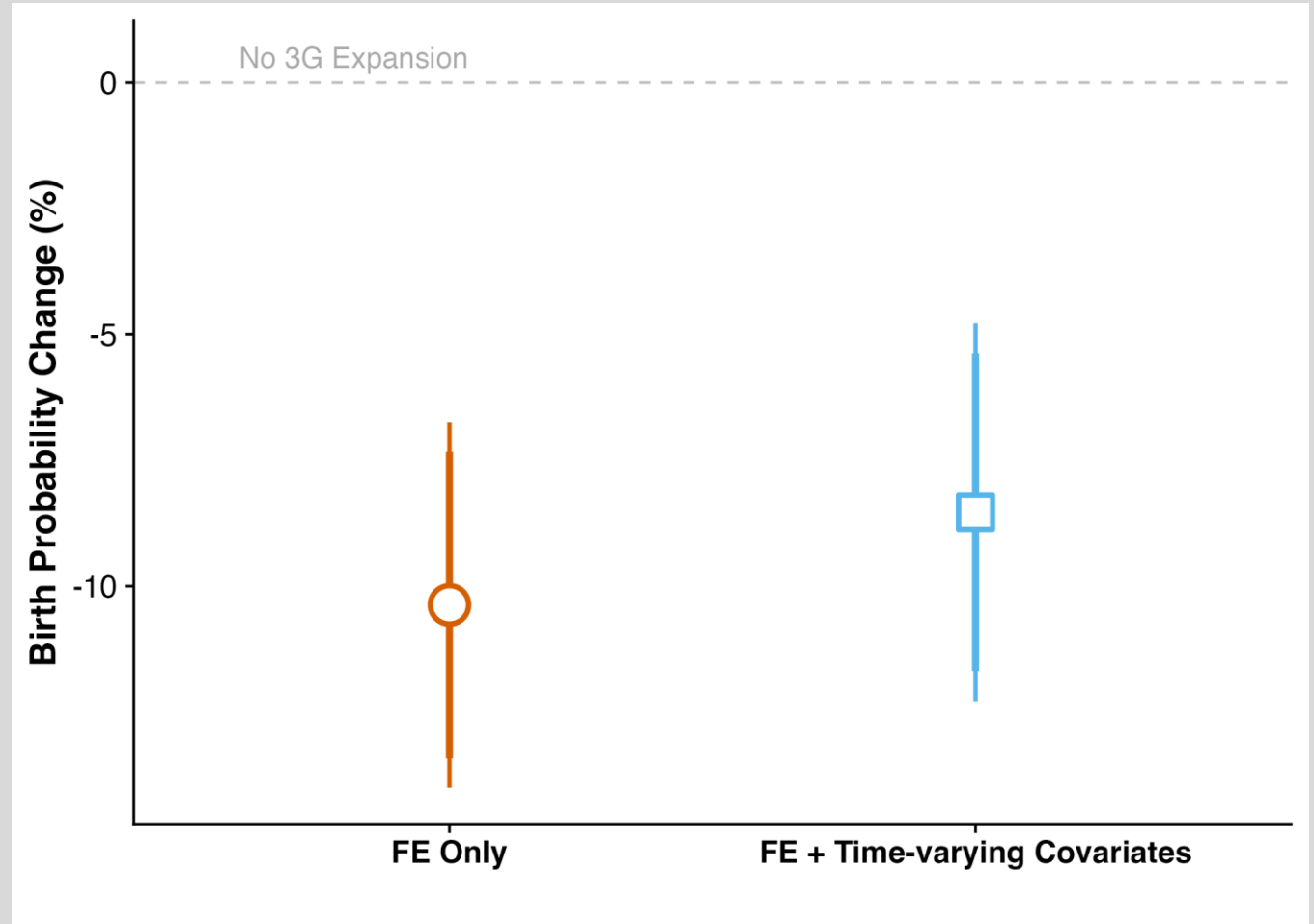
- We study the impact of 3G expansion using a two-way fixed effects (TWFE), with cluster and year FE, individual-level controls, and time-varying cluster controls (2G, GDP, nightlights, temperature, rain)

$$LB_{ict} = \beta_0 + \underbrace{\beta_1 3G_{ct}}_{\text{3G coverage intensity}} + \underbrace{\gamma_c}_{\text{Cluster FE}} + \underbrace{\delta_t}_{\text{Year FE}} + \underbrace{\beta X_i}_{\text{Controls}} + \epsilon_{ict}$$

↓
a unitary increase in 3G coverage in a cluster i.e.
moving from 0% of population covered to 100%
population covered by 3G, lagged by 1 year

Results

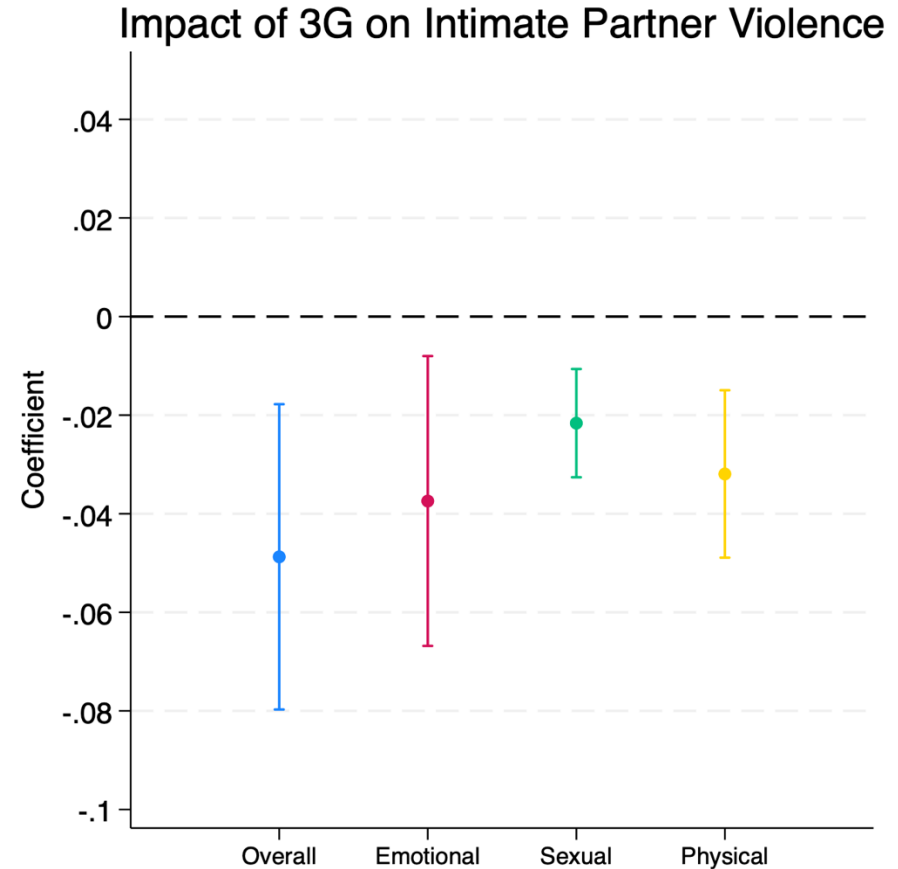
We observe a **7% reduction in the birth probability** (with time-varying controls + cluster and year FEs) with full 3G coverage



Results

In a shift from 0% to 100% population covered by 3G in cluster, women faced:

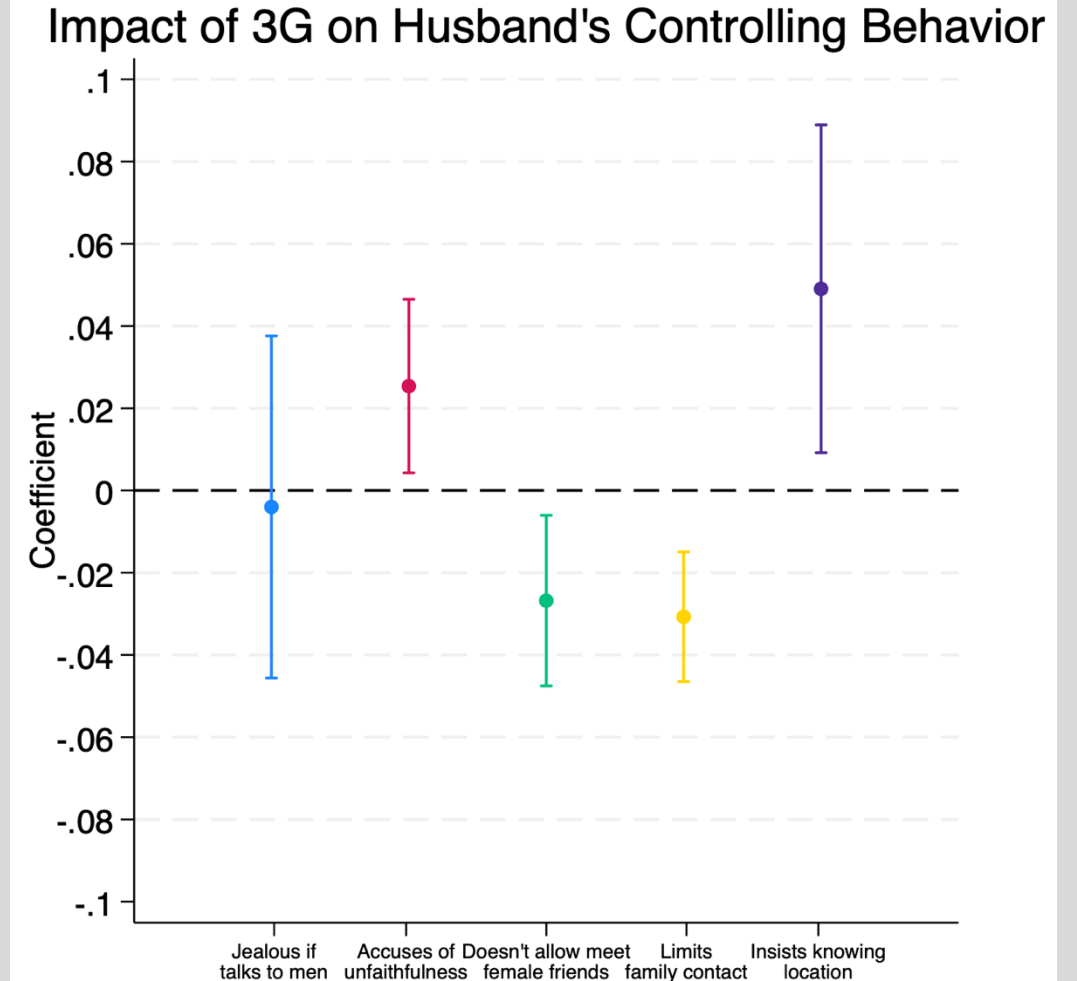
- **3.7pp lower emotional violence** (19% reduction from mean level),
- **2.1pp lower sexual violence** (56% reduction from mean level)
- **3.1pp lower physical violence** (26% reduction from mean level)



Results

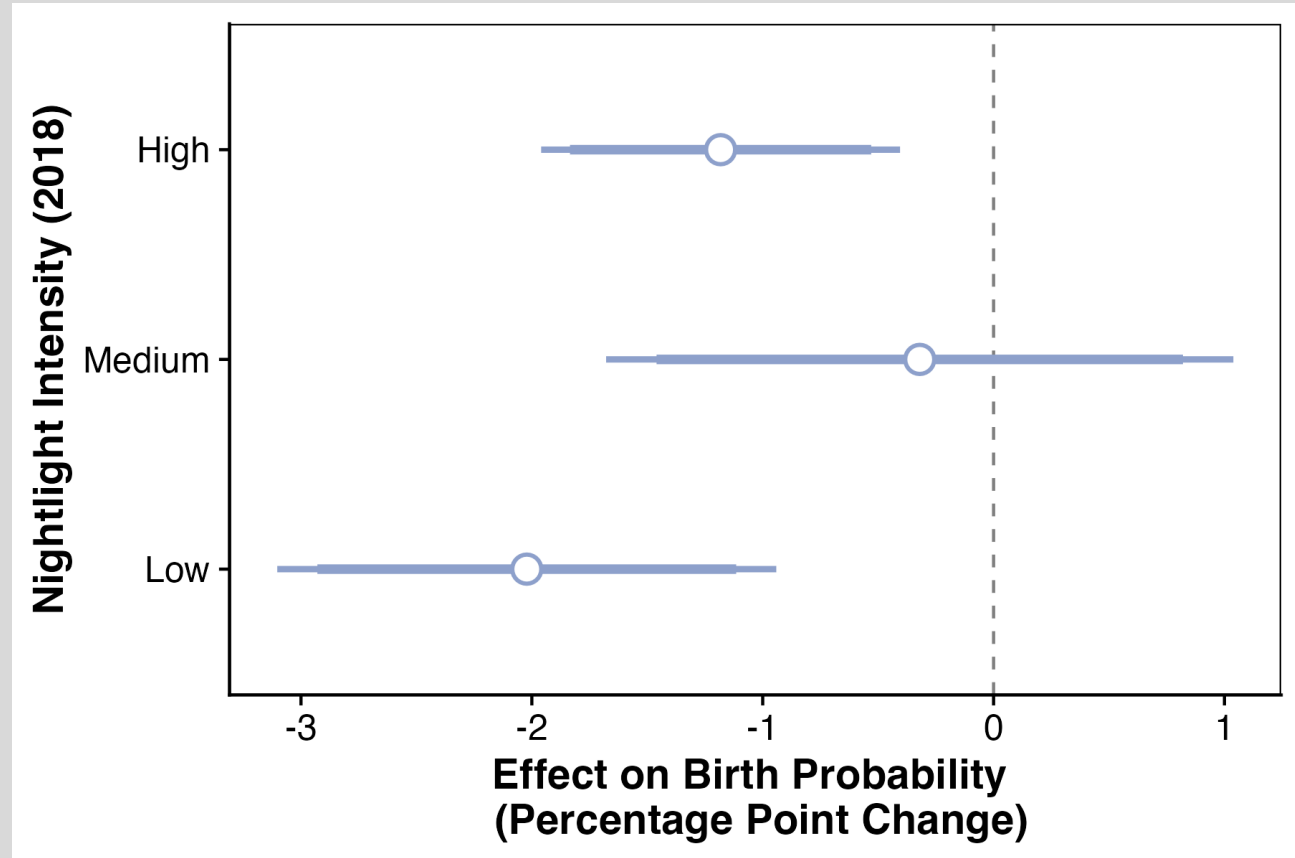
Shifts in men's controlling behaviors may reflect both empowerment and backlash dynamics:

While some behaviors—like limiting social contact—declined, others (e.g., accusations of infidelity, monitoring wives' locations) increased.



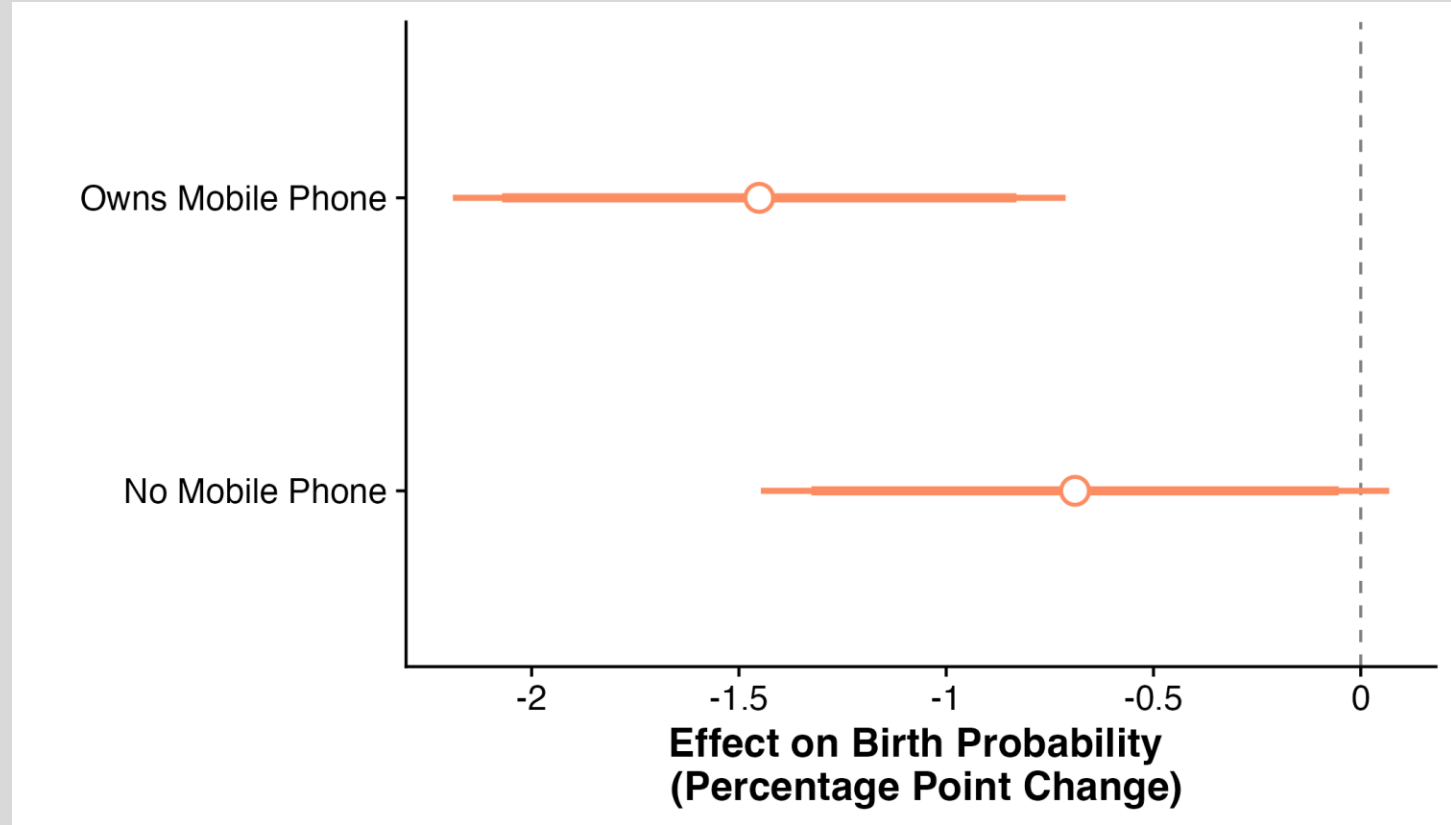
Heterogeneity by Development Level

Larger reductions in areas at lowest and highest levels of local development, as proxied by nightlights.



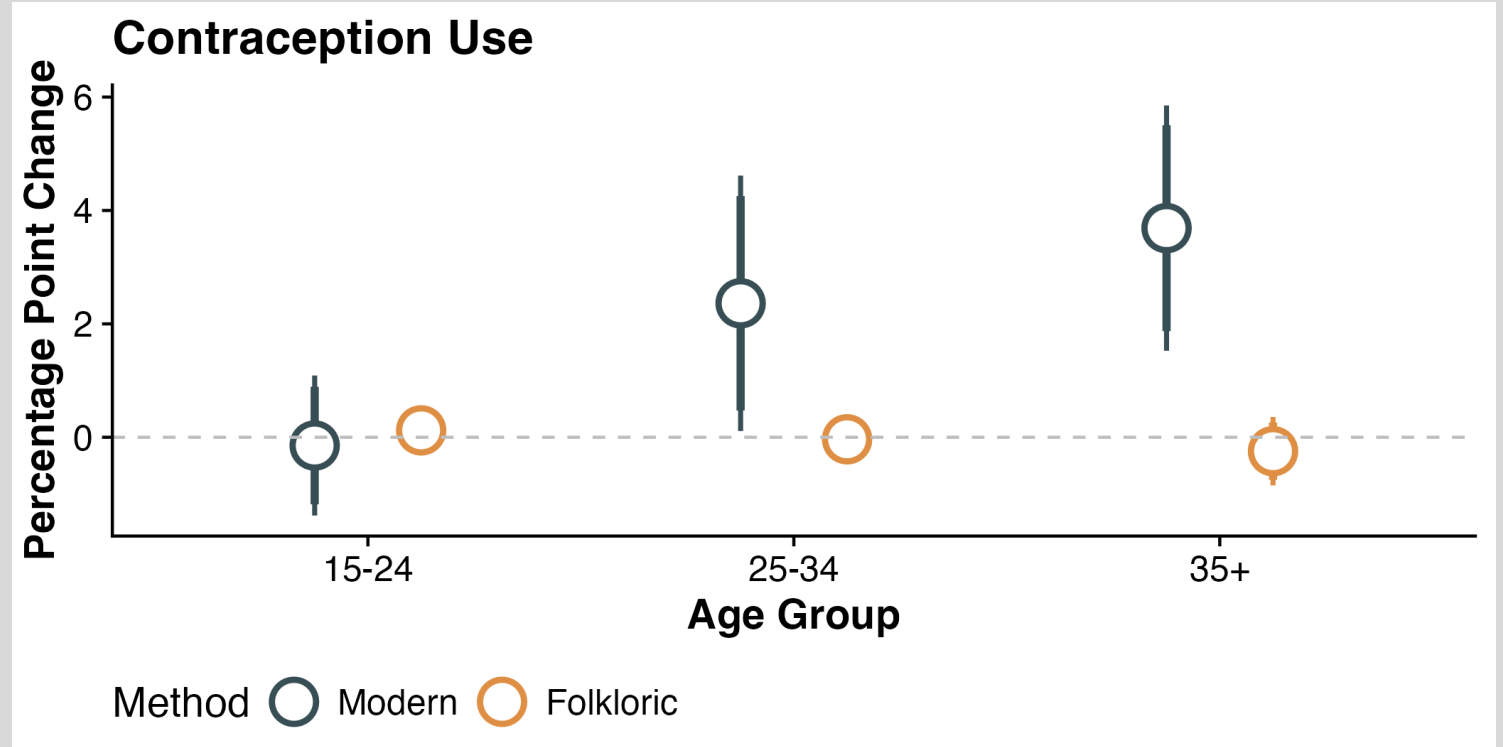
Heterogeneity by Mobile Phone Ownership

Larger reductions among those who own mobile phones



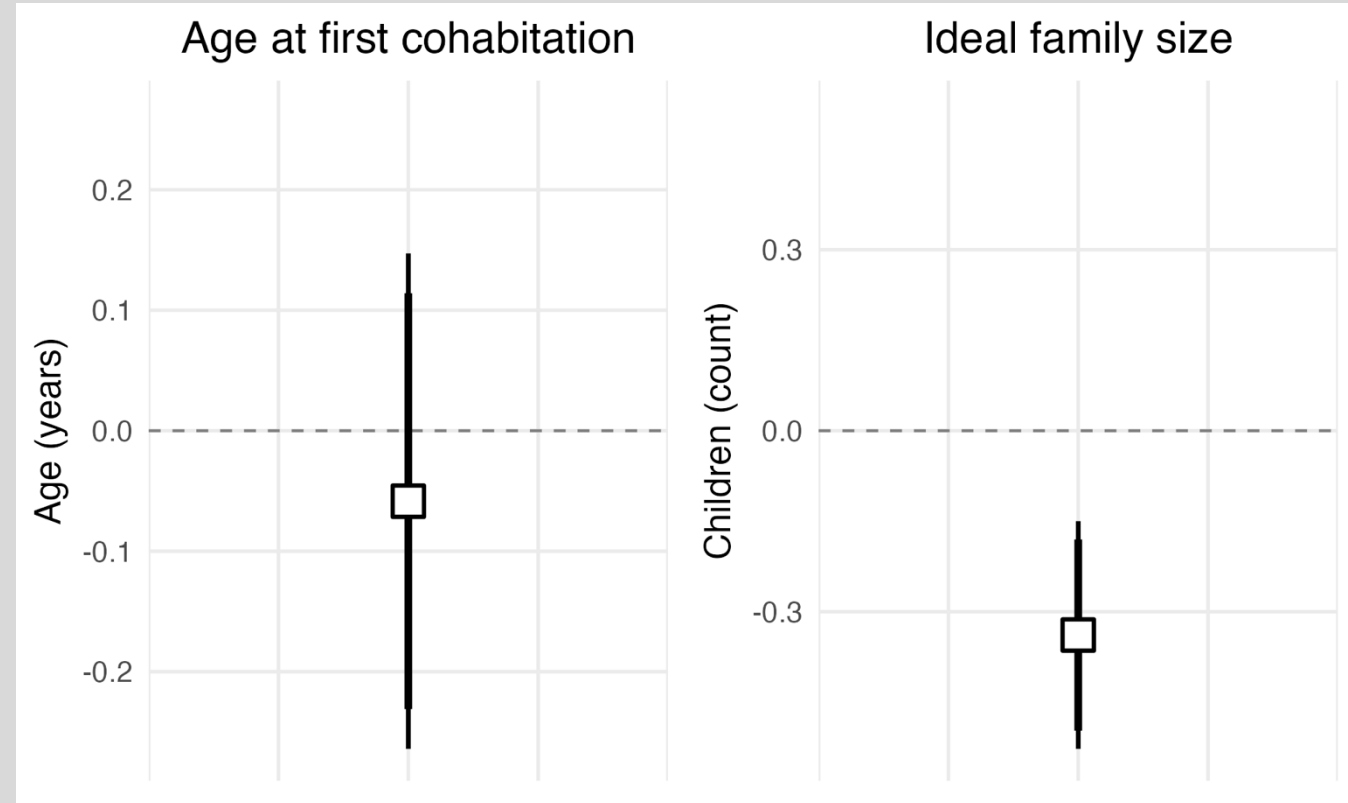
Mechanisms: Ideational

Evidence for **changes in modern contraceptive knowledge and use**



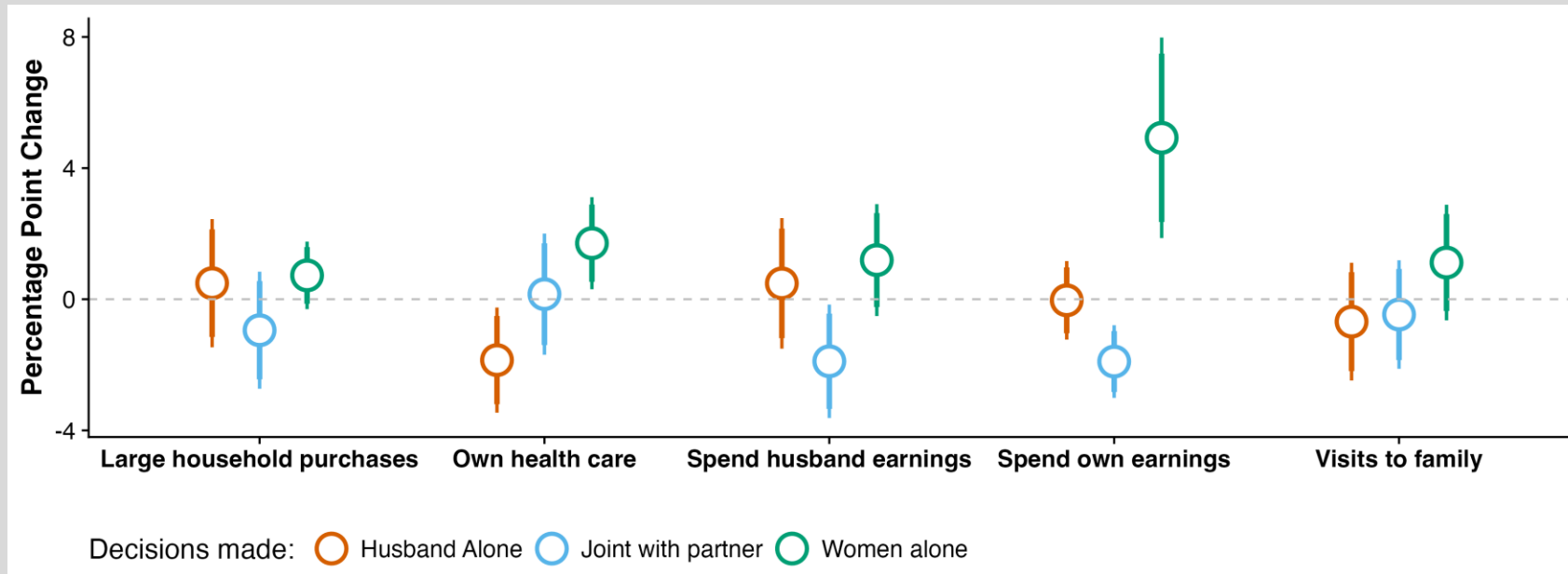
Mechanisms: Ideational

Reductions in and reductions in **ideal family size** but not in age at first cohabitation



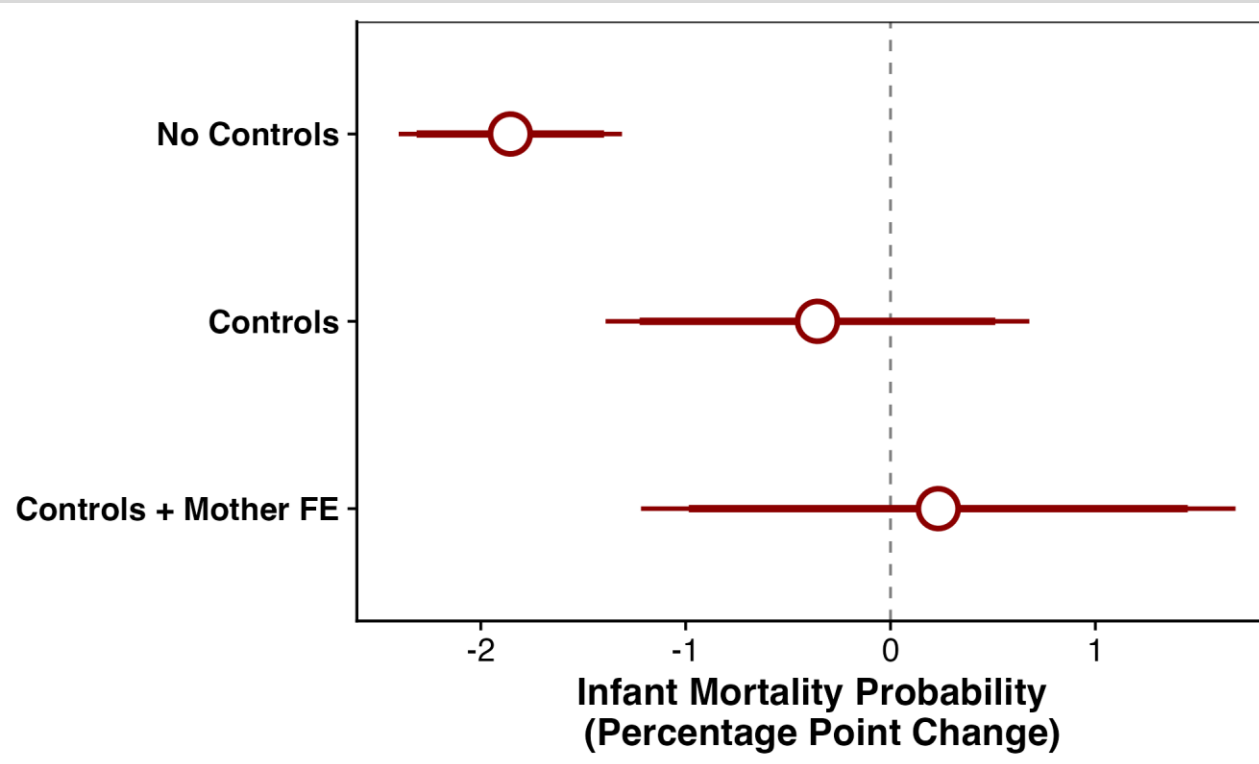
Mechanisms: Autonomy

Evidence for **increases in health and financial independent decision-making** for women



Mechanisms: Demographic

No statistically significant changes in demographic mechanisms linked to **infant mortality** or in **age at first cohabitation**



Summary (1)

- The digital revolution has implications for **measuring and understanding** gender inequalities
- Social media data streams integrated with conventional sources of population data, e.g. from surveys, enable better temporal and geographical coverage of gender inequality indicators
- Empirical evidence for impacts of **social media networking and visibility** on professional outcomes, with potentially larger payoffs for women
 - But offline inequalities are also reinforced online
- **Digital information and capabilities** can positively impact women's status and empowerment; but also generate backlash and controlling behaviours within households with power shifts

Summary (2)

- Digital gender gaps are an important dimension of population inequality: our estimates show women are ~20% likely to use the internet in LMICs (2025) → not AI ready
- How do technology impacts shift as tools and capabilities shift?
- To understand the impacts of digital, we need to integrate both 'old' and 'new' data within a **diversifying data ecosystem**

Thank you!

ridhi.kashyap@nuffield.ox.ac.uk

Learn more about us at:

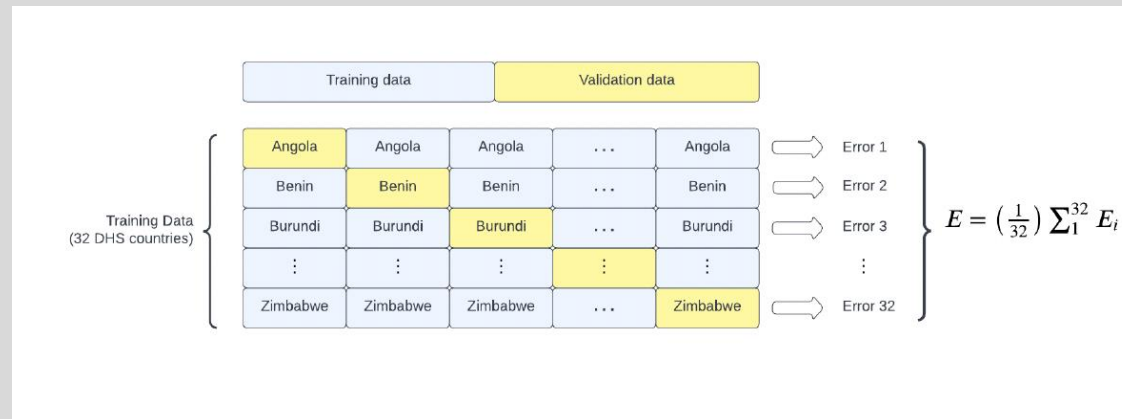
www.digitalgendergaps.org

We are grateful to funding from the Gates Foundation (INV-045370) and Leverhulme Trust for the Leverhulme Centre for Demographic Science (Grant RC-2018-003).

DGG Team and Collaborators: Casey Breen, Hannah Gardner, Ghazal Kalhor, Till Koebe, Douglas Leasure, Jiaxuan Li, Prashant Poddar, Valentina Rotondi, Florianne Verkroost, Ingmar Weber, Jiani Yan, Xinyi Zhao

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Subnational Gaps

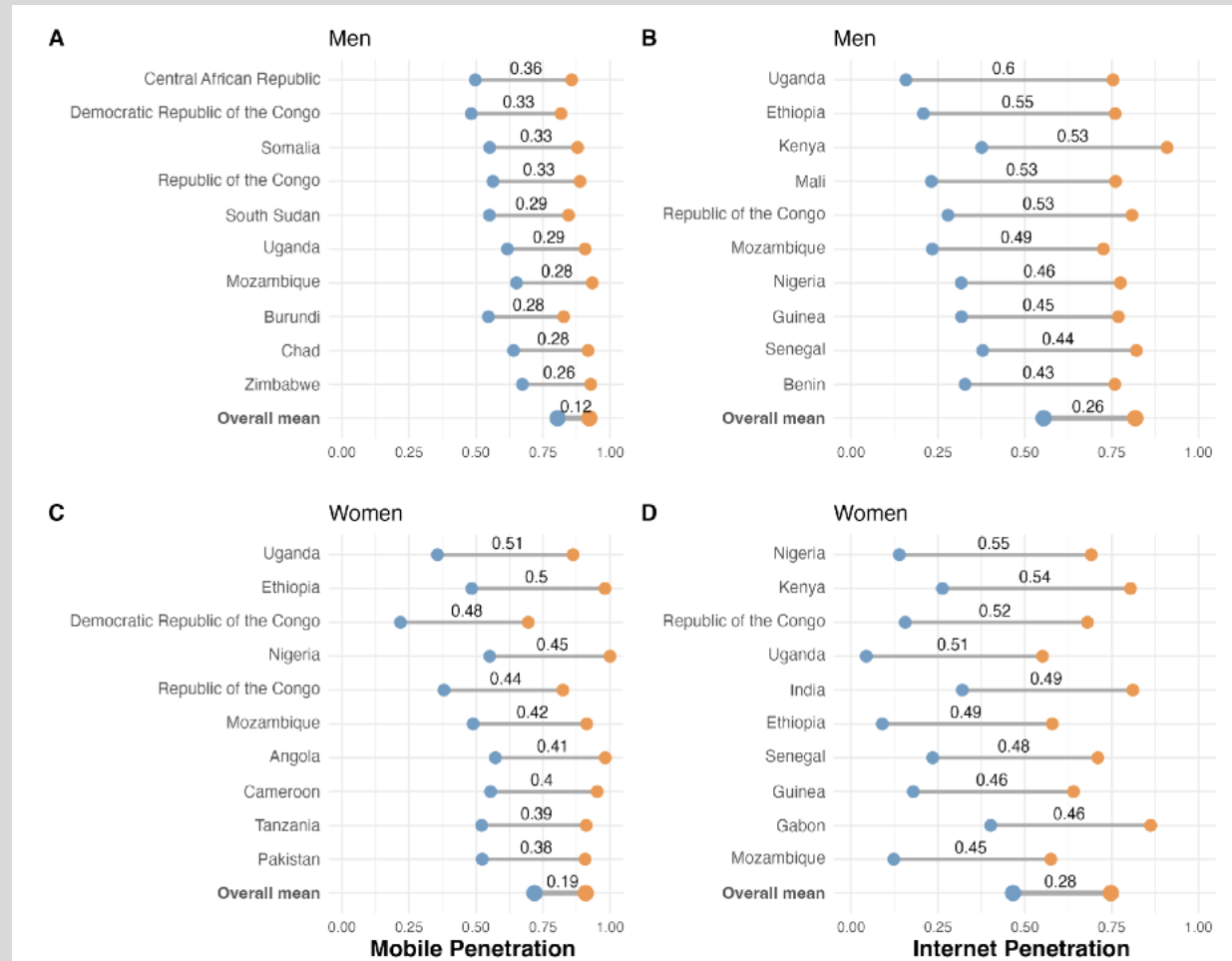


Figure 5: The top 10 countries with the largest gaps between their lowest and highest subnational unit by gender and digital indicator, organized in descending order by gap size. The bottom bar shows the average top-bottom subnational disparity across all countries.

Decomposing inequality

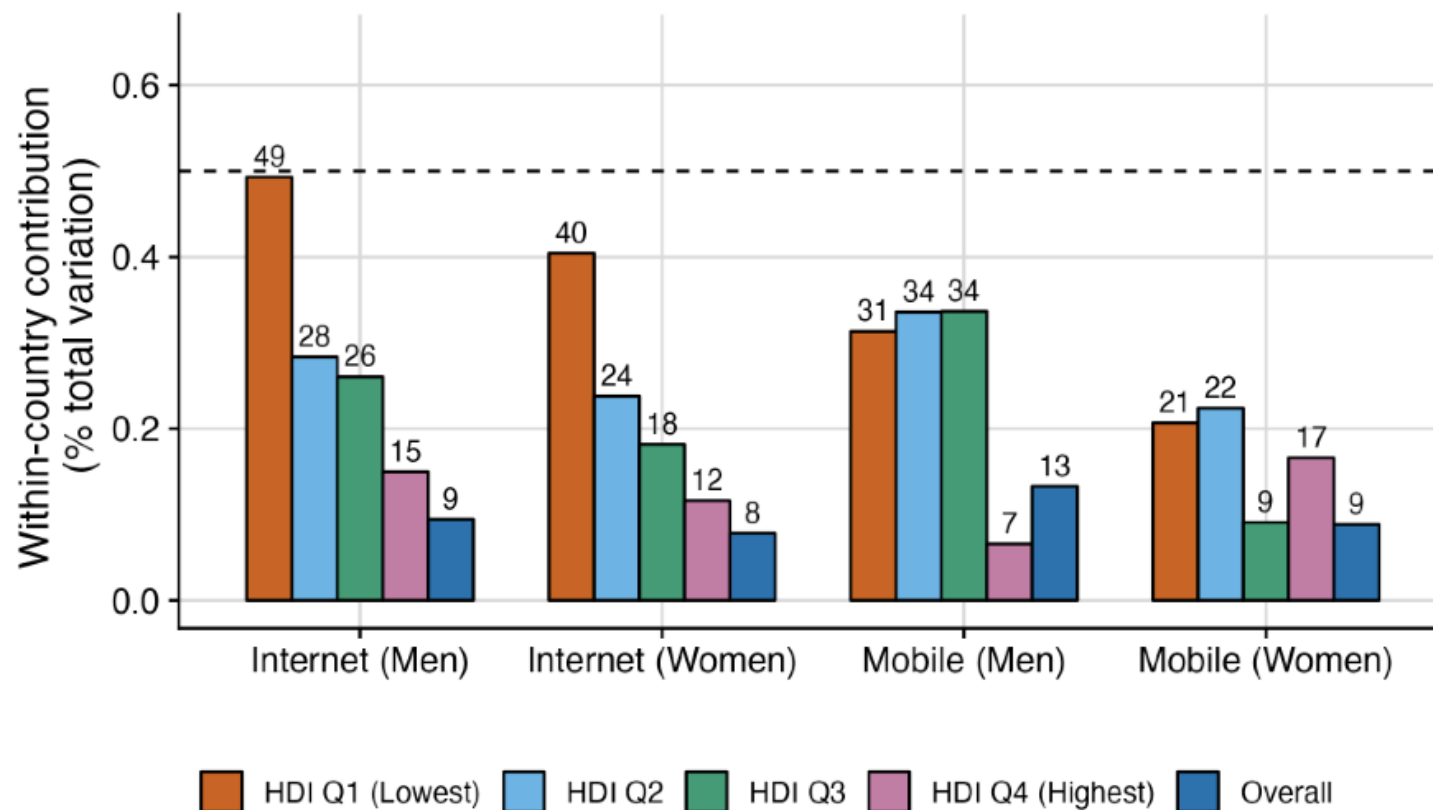
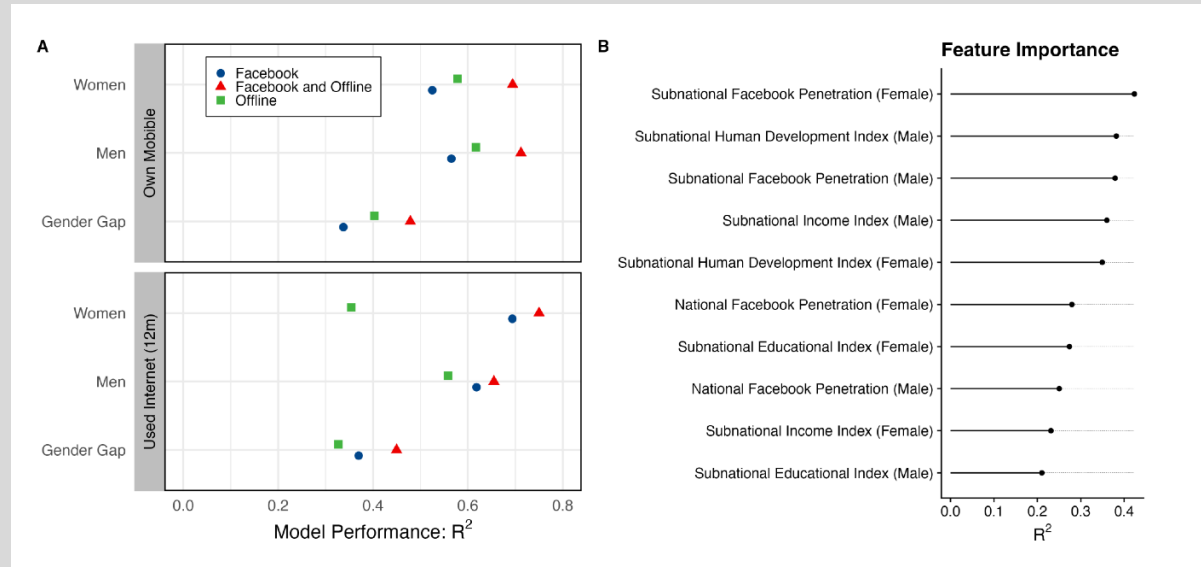


Figure 4: The relative contribution of within-country variation in internet and mobile adoption to total variation by human development index (HDI) quantile. Dashed line represents equal contribution of within-country and between-country variation to overall inequality.

Insights from Predictive Models

- Combining Facebook features with development indicators at the subnational levels yields best performing models
- Key non-social media features for predicting digital adoption include human development index; educational factors matter more for women's digital adoption



Expanding across social media platforms and data types

- Instagram, Snapchat, TikTok (examining digital behaviours of different population sub-groups, e.g. adolescents)
- Google AdWords (model stability/coverage)
- LinkedIn ad audience estimates (differential types of platform and uses)
- Social connectedness between regions as captured on social media friendships (Meta's Social Connectedness Index): measures of social networks and social capital and its link with sexual and reproductive health inequalities

Other Applications:

Gender Gaps in Online Professional Networking

**Use a matched audience** (optional)
Custom targeting options to reach your website visitors, contacts, and target accounts.

Select

Target by

the audience below

What location do you want to target? (required)

include

Start typing a country, state, city, or town...

See full list

include

Netherlands

☒ Target people who permanently live or work in the selected location(s).
Deliver ads to people who reside in the selected **location(s)** and are not recent visitors

What gender do you want to target?

☐ All

☐ Female

☒ Male

Select specific targeting criteria to zero in on your ideal audience:

Company name

Company industry

Company size

Job title

Job function

Job seniority

Member schools

Fields of study

Degrees

Member skills

Your estimated target audience

4,500,000+ LinkedIn members

Learn more

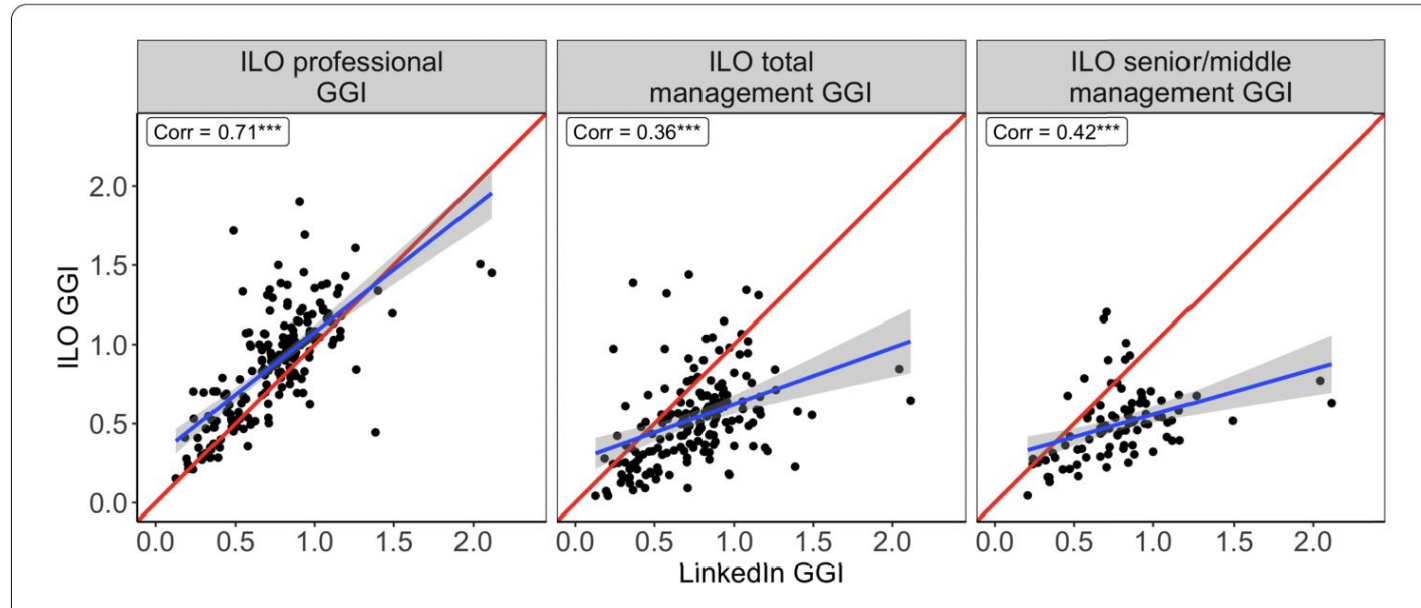
Netherlands

Male

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Gender Gaps in Online Professional Networking



Women in management or senior/middle management roles are *overrepresented* on LinkedIn relative to labour force: online networking may provide greater benefits

LinkedIn Gender Gap Index (female-to-male ratio) and International Labour Organisation (ILO) Gender Gap Index across countries

- We use different model performance metrics, incl. R2, MAE

$$\begin{aligned} R^2 &= 1 - \frac{SS_{res}}{SS_{tot}} \\ &= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \end{aligned}$$

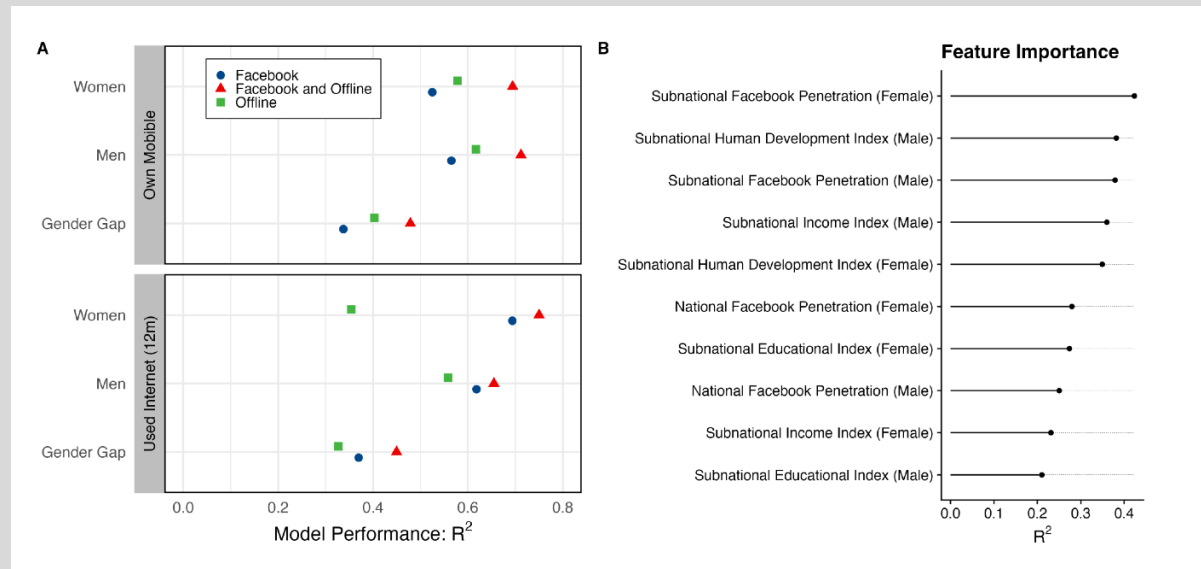
- 1 = perfect prediction
- 0 = mean

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|.$$

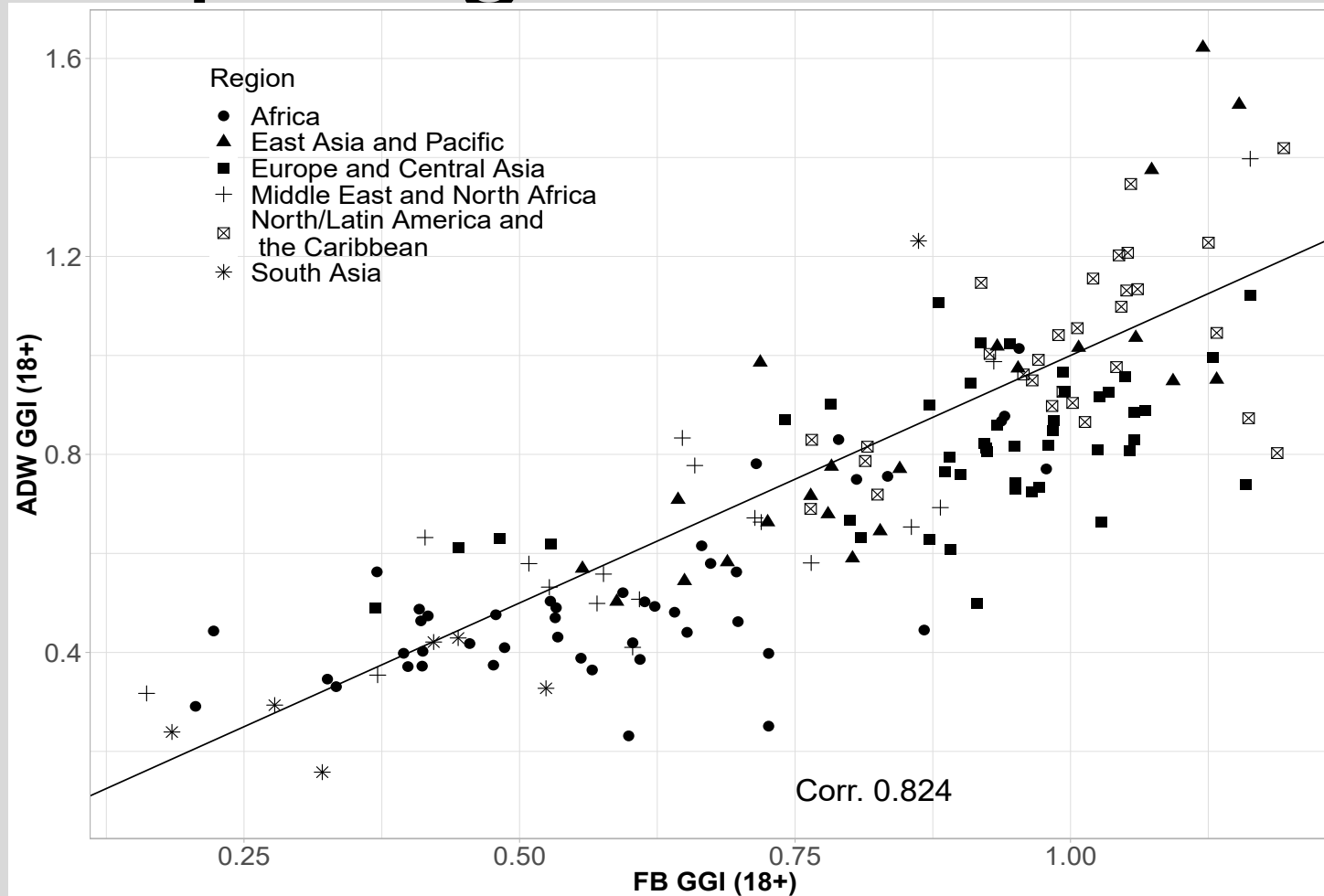
MAE provides an absolute measure of the average prediction error in dependent variable's units; lower values mean lower error

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- Key non-social media features for predicting digital adoption include human development index; educational factors matter more for women's digital adoption

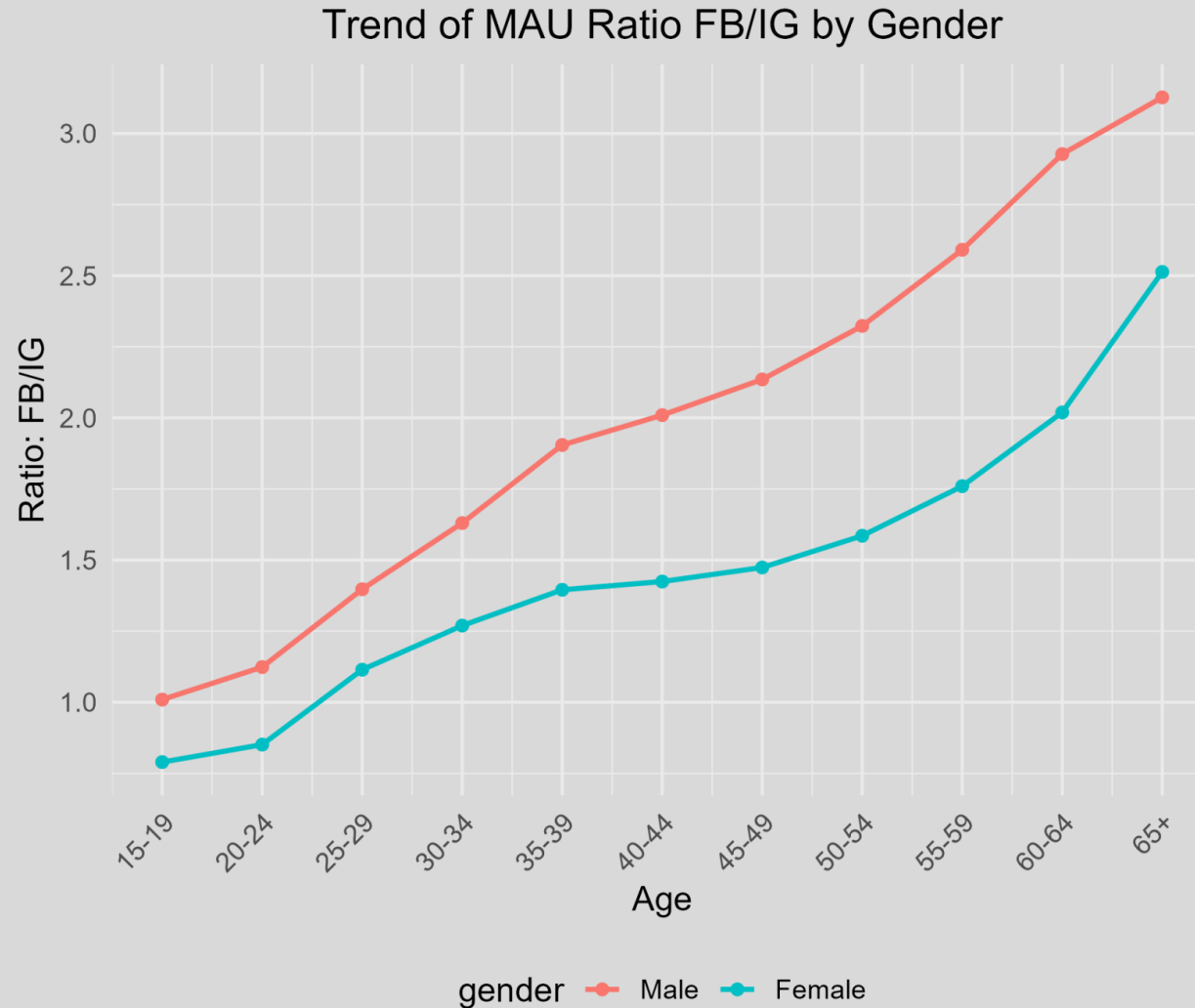


Comparing FB with AdWords



- FB is independently better at predicting Internet GGI at national level
 - But, best-performing models combine FB and AdWords
- Good proxies of low-level digital skills (e.g copy-paste; file transfer)

Instagram and FB: age/gender



Relative use of Facebook versus Instagram:

Monthly active users (MAU) on Facebook v Instagram (ratio of FB/IG) by gender and age-group.